Thesis draft

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Chapter 1

Introduction

The central theme of this thesis is the interplay between quantum field theory and cosmology during the early universe. The key observation is this: if matter is described by a quantum field with a nonvanishing potential, there is a natural mechanism for generating an effective cosmological constant in the early universe, which drives a phase of exponential expansion known as inflation. This in turn provides an elegant explanation for the homogeneity and flatness of the observed universe.

We work within the framework of grand unified theories (GUTs), where gravity is described classically by a Friedmann-Robertson-Walker (FRW) metric and matter is described by quantum field theory. At $T = 0$, the gauge symmetry of the GUT is spontaneously broken by a nonzero vacuum expectation value of a scalar Higgs field. At high temperature the symmetry is restored, but at the cost of generating a large cosmological constant. As the universe cools, it passes through one or more phase transitions.

The reason for describing matter using quantum fields is well-motivated: for times earlier than 10^{-6} s after the Big Bang, the thermal energy of particles exceeds 1 GeV, and at these energy scales particle interactions are correctly described by QFT, as confirmed by particle accelerator experiments. An ideal gas approximation is simply inadequate at such temperatures.

The central tool of this thesis is the *effective potential* $V_{\text{eff}}(\bar{\varphi})$. To understand why it is the right object, consider a scalar field φ coupled to other fields A_μ. The classical potential $V(\varphi)$ gives the energy of a homogeneous stationary state with $\varphi(x) = \varphi$ and $A_\mu = 0$. The effective potential $V_{\text{eff}}(\bar{\varphi})$ is the minimum expectation value of the energy density in the class of all homogeneous stationary states with expectation value $\bar{\varphi}$. This is the main reason it is of interest: the value of $\bar{\varphi}$ in the quantum state at any given time minimizes $V_{\text{eff}}(\bar{\varphi})$ at that time.

A large positive potential $V(\phi)$ in some range of field values, with a quantum state that is homogeneous and stationary in that range, causes the potential contribution to the energy-momentum tensor to dominate, acting as an effective cosmological constant in Einstein's equations. Finding the state of the quantum field as the universe evolves therefore reduces to finding the minimum of the effective potential.

The effective potential also governs the equation of motion of the expectation value $\bar{\varphi}$,

$$\frac{\partial^2 \bar{\varphi}}{\partial t^2} - \nabla^2 \bar{\varphi} = -\frac{\partial}{\partial \bar{\varphi}} V_{\text{eff}}(\bar{\varphi}), \tag{1.1}$$

which is the leading contribution in an expansion about a homogeneous and stationary state. The effective potential thus replaces the classical potential in the quantum equation of motion, as we will show explicitly.

Two reasons motivate the use of grand unified gauge field theories in this context. First, particle accelerator experiments suggest that fundamental interactions are described by gauge theories at high energies. Second, the cosmological constant Λ vanishes today, $\Lambda/m_{\rm Pl}^2 < 10^{-110}$, so for Λ to have been large in the early universe, a phase transition from $\langle V(\phi)\rangle \neq 0$ to $\langle V(\phi)\rangle = 0$ must have occurred. GUTs have such phase transitions built in.

The temperature dependence of the effective potential is crucial here. Since the effective action is the free energy of the quantum state, the effective potential is the free energy density for homogeneous states, and statistical mechanics tells us that the free energy is temperature dependent. Concretely, finite-temperature corrections generate a mass term proportional to $\varphi^2 T^2$, which restores the symmetry at high temperature: even if $\varphi = 0$ is not the minimum at $T = 0$, it becomes so at sufficiently high T. The high-T and low-T ground states are therefore different.

In GUT models with a Higgs mechanism, the gauge symmetry is broken by a nonvanishing VEV of φ. Finite-temperature corrections to the effective potential restore the symmetry above a critical temperature T_c, set by the scale of symmetry breaking. As the universe cools below T_c, a new absolute minimum of $V_{\rm eff}(\varphi)$ appears at $\varphi = \sigma \neq 0$ and a phase transition occurs. Since Λ must be fine-tuned to zero today, $V_{\rm eff}(\sigma) = 0$, it follows that $V_{\rm eff}(0) > 0$ in the high-T phase. GUTs with Higgs symmetry breaking therefore automatically generate a period of inflation preceding the phase transition.

In the semiclassical approximation, the decay of the false vacuum proceeds via *bubble nucleation*: the true stable phase, with vanishing Λ, nucleates as a bubble and expands at the speed of light into the surrounding sea of false vacuum. The analysis of this process is one of the main topics of this thesis.

In curved spacetime, new effects arise. Unlike flat spacetime, where the Poincaré-invariant Minkowski vacuum is agreed upon by all inertial observers, general relativity is diffeomorphism invariant and no frame is globally preferred. As a consequence, there is no unique vacuum state in a general curved spacetime. The most celebrated manifestation of this is Hawking radiation: an observer with a particle detector in certain spacetimes will, at late times, measure a nonvanishing thermal flux of particles in a state that was initially empty. We discuss these effects in the final part of the thesis.

The thesis is organized as follows. Part I develops the effective action formalism at zero temperature, building from the generating functionals to the one-loop effective potential and its application to gauge theories. Part II extends this to finite temperature using the imaginary-time formalism. Part III treats cosmological phase transitions, including the thermal history of the early universe and the parameters characterizing a first-order transition. Part IV analyzes false vacuum decay and bubble nucleation via the bounce solution. Part V discusses quantum field theory in curved spacetime, culminating in Hawking radiation.

Part I

Effective Action and Effective Potential

Chapter 2

General Formalism of the Effective Action

The central object of this thesis is the *effective action* $\Gamma[\varphi_c]$, a functional that encodes the full quantum dynamics of a field theory in a single object. Before defining it, we build up to it systematically: starting from the generating functional $Z[J]$ for all Green functions, passing through the connected generating functional $W[J]$, and finally arriving at $\Gamma[\varphi_c]$ via a Legendre transform. The payoff is that the equation of motion derived from Γ governs the quantum-corrected vacuum structure of the theory, making it the natural tool for studying spontaneous symmetry breaking and phase transitions.

2.1 Generating Functionals

Rather than computing Green functions diagram by diagram, all of them, full, connected, and one-particle-irreducible, can be extracted from a small number of functionals by simple functional differentiation.

Coupling the field $\varphi(x)$ linearly to an arbitrary c-number source $J(x)$,

$$\mathcal{L}(\varphi, \partial_\mu \varphi) \;\longrightarrow\; \mathcal{L} + J(x)\,\varphi(x), \tag{2.1}$$

the generating functional for the full Green functions is defined as the vacuum-to-vacuum transition amplitude in the presence of this source,

$$Z[J] = \langle 0|T \exp\left(i \int J(x)\,\varphi(x)\,d^4x \right) |0\rangle, \tag{2.2}$$

where $|0\rangle$ is the physical vacuum and $\varphi(x)$ is the Heisenberg-picture field, both defined at $J = 0$. Expanding in powers of J makes the content of this definition explicit,

$$Z[J] = \sum_{n=0}^{\infty} \frac{i^n}{n!} \int d^4x_1 \cdots d^4x_n \; J(x_1) \cdots J(x_n)\; G_n(x_1, \ldots, x_n), \tag{2.3}$$

where $G_n(x_1, \ldots, x_n) \equiv \langle 0|T[\varphi(x_1) \cdots \varphi(x_n)]|0\rangle$ is the full n-point Green function. Each G_n is recovered by n functional derivatives of $Z[J]$ with respect to J, evaluated at $J = 0$. Equivalently, $Z[J]$ admits a path-integral representation,

$$Z[J] = N \int [\mathcal{D}\varphi]\; \exp\left(i \int d^4x \left[\mathcal{L}[\varphi(x)] + J(x)\,\varphi(x) \right] \right), \tag{2.4}$$

where N is fixed by $Z[0] = 1$. The two representations (2.2) and (2.4) describe the same object and will be used interchangeably throughout.

The full Green functions G_n contain disconnected contributions, products of lower-point functions, that carry no new dynamical information. The *connected* Green functions G_n^c, which isolate only the irreducible correlations, are generated by passing to the logarithm of $Z[J]$,

$$iW[J] = \ln Z[J], \tag{2.5}$$

the proof that this logarithm precisely removes all disconnected pieces is given in Appendix A. Since $Z[J]$ is the vacuum-to-vacuum amplitude, Eq. (2.5) means

$$e^{iW[J]} = \langle 0^+ | 0^- \rangle_J, \tag{2.6}$$

where $|0^\pm\rangle$ are the vacuum states in the far future and far past, so $W[J]$ is the *vacuum persistence amplitude* in the presence of the external source. This interpretation will be central when we discuss the metastability of the false vacuum in Chapter 9.

The first functional derivative of $W[J]$ defines the *classical field*,

$$\varphi_c(x) \equiv \frac{\delta W}{\delta J(x)} = \left[\frac{\langle 0^+ | \varphi(x) | 0^- \rangle}{\langle 0^+ | 0^- \rangle} \right]_J, \tag{2.7}$$

the source-dependent vacuum expectation value of $\varphi(x)$, which will serve as the argument of the effective action.

2.2 The Effective Action and the Effective Potential

Given $W[J]$ and the classical field φ_c via (2.7), we trade the source J for φ_c as the fundamental variable via the Legendre transform,

$$\Gamma[\varphi_c] = W[J] - \int d^4x \, J(x) \, \varphi_c(x), \tag{2.8}$$

where J on the right-hand side is understood as a functional of φ_c obtained by inverting (2.7). Differentiating with respect to φ_c and using (2.7) gives

$$\frac{\delta \Gamma}{\delta \varphi_c(x)} = -J(x). \tag{2.9}$$

In the physical situation $J = 0$, this becomes the quantum equation of motion,

$$\frac{\delta \Gamma}{\delta \varphi_c(x)} = 0, \tag{2.10}$$

the exact analogue of $\delta S / \delta \varphi = 0$ for the classical action, but now with all quantum corrections included. The vacua of the quantum theory are precisely the solutions to (2.10).

Expanding $\Gamma[\varphi_c]$ in powers of φ_c,

$$\Gamma[\varphi_c] = \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \cdots d^4x_n \, \Gamma^{(n)}(x_1, \ldots, x_n) \, \varphi_c(x_1) \cdots \varphi_c(x_n), \tag{2.11}$$

the coefficients $\Gamma^{(n)}$ are the one-particle-irreducible (1PI) Green functions, or proper vertices, the sum of all Feynman diagrams with n external lines that cannot be disconnected by cutting a single internal propagator, with no external propagators attached.

For a translationally invariant configuration $\varphi_c = \text{const}$, the effective action takes the form $\Gamma = -\Omega\, V(\varphi_c)$, where Ω is the four-volume. More generally, for slowly-varying fields, Γ admits a derivative expansion,

$$\Gamma[\varphi_c] = \int d^4x \left[-V(\varphi_c) + \frac{1}{2}(\partial_\mu \varphi_c)^2\, Z(\varphi_c) + \cdots \right], \tag{2.12}$$

where $V(\varphi_c)$ is the *effective potential*, an ordinary function rather than a functional. Comparing with (2.11), the n-th derivative of V equals the sum of all 1PI diagrams with n external legs at zero external momenta, so the effective potential encodes the full quantum-corrected vacuum structure of the theory. The physical mass and coupling are read off from it via

$$\mu^2 = \left.\frac{d^2V}{d\varphi_c^2}\right|_{\varphi_c=0}, \qquad \lambda = \left.\frac{d^4V}{d\varphi_c^4}\right|_{\varphi_c=0}. \tag{2.13}$$

2.3 Spontaneous Symmetry Breaking

The effective potential is the natural language for spontaneous symmetry breaking. Suppose $\mathcal{L}$ has a discrete internal symmetry, e.g. $\varphi \to -\varphi$. This symmetry is spontaneously broken when the quantum equation of motion (2.10) admits a solution with a nonzero, spatially uniform vacuum expectation value,

$$\left.\frac{dV}{d\varphi_c}\right|_{\varphi_c=\langle\varphi\rangle} = 0, \qquad \langle\varphi\rangle \neq 0, \tag{2.14}$$

even in the absence of any external source. Stability requires this stationary point to be a minimum of V.

Two phases are possible. In the symmetric phase, $\langle\varphi\rangle = 0$ and the $\varphi \to -\varphi$ symmetry is intact. In the broken phase, $\langle\varphi\rangle = v \neq 0$ and the symmetry is hidden. The physical spectrum in the broken phase is described by the shifted field $\varphi' = \varphi - v$, and the renormalization conditions (2.13) are displaced to the broken minimum,

$$\mu^2 = \left.\frac{d^2V}{d\varphi_c^2}\right|_{\varphi_c=v}, \qquad \lambda = \left.\frac{d^4V}{d\varphi_c^4}\right|_{\varphi_c=v}. \tag{2.15}$$

With the formalism in place, we turn in the next chapter to an explicit one-loop computation of the effective potential, first in a simple scalar theory and then in the physically relevant case of scalar electrodynamics.

Chapter 3

Physical Meaning of the Effective Action and Effective Potential

We have defined $\Gamma[\varphi_c]$ and $V_{\text{eff}}(\bar{\varphi})$ formally in the previous chapter. Here we give them concrete physical meaning.

3.1 The Effective Potential as a Generator of 1PI Graphs

For a constant classical field $\bar{\varphi}$, inserting the Fourier representation of $\Gamma^{(n)}$ into the expansion (2.11) gives

$$\Gamma(\bar{\varphi}) = \sum_{n=2}^{\infty} \frac{1}{n!}\, \bar{\varphi}^n \int \prod_{i=1}^{n} d^4x_i\, \Gamma^{(n)}(x_1, \ldots, x_n) \tag{3.1}$$

$$= \sum_{n=2}^{\infty} \frac{1}{n!}\, \bar{\varphi}^n \int \prod_{i=1}^{n} \left(d^4x_i\, \frac{d^4k_i}{(2\pi)^4}\, e^{ik_i x_i} \right) \tilde{\Gamma}^{(n)}(k_1, \ldots, k_n) \cdot (2\pi)^4\, \delta^{(4)}\left(\sum_{i=1}^{n} k_i \right), \tag{3.2}$$

where $\int d^4x\, e^{ikx} = (2\pi)^4 \delta^{(4)}(k)$ was used. Setting $k = 0$ gives $\int d^4x = (2\pi)^4 \delta^{(4)}(0)$, so that

$$\Gamma(\bar{\varphi}) = \sum_{n=2}^{\infty} \frac{1}{n!}\, \bar{\varphi}^n \int d^4x\, \tilde{\Gamma}^{(n)}(0, \ldots, 0). \tag{3.3}$$

Comparing this with $\Gamma(\bar{\varphi}) = \int d^4x\, [-V_{\text{eff}}(\bar{\varphi})]$ gives

$$V_{\text{eff}}(\bar{\varphi}) = -\sum_{n=2}^{\infty} \frac{1}{n!}\, \bar{\varphi}^n\, \tilde{\Gamma}^{(n)}(0, \ldots, 0). \tag{3.4}$$

Thus $V_{\text{eff}}(\bar{\varphi})$ is the generating functional for 1PI graphs at vanishing external momenta, computed by summing all such graphs and inserting a factor $\bar{\varphi}$ for each external line.

To lowest order in $\hbar$, only tree-level graphs contribute. For a Lagrangian of the form

$$\mathcal{L}(\varphi, \partial_\mu \varphi) = \frac{1}{2}\, \partial_\mu \varphi\, \partial^\mu \varphi - U(\varphi), \tag{3.5}$$

each term $g_i \varphi^i / i!$ in $U(\varphi)$ contributes exactly one tree-level 1PI graph, since all tree graphs with more than one vertex are not 1PI. Each term gives a contribution $-g_i \bar{\varphi}^i$, where $-g_i$

is the vertex factor and $\bar{\varphi}^i$ comes from the i external lines. The minus signs cancel and one recovers

$$V_{\text{eff}}(\bar{\varphi}) = U(\bar{\varphi}) + \mathcal{O}(\hbar), \tag{3.6}$$

confirming that the classical potential is the leading-order approximation to the effective potential.

3.2 The Effective Potential as a Minimum Energy Density

A deeper interpretation follows from the variational principle.

Theorem 1. *$V_{\text{eff}}(\bar{\varphi})$ is the minimum of the energy density expectation value over all normalized states $|a\rangle$ satisfying $\langle a|\varphi|a\rangle = \bar{\varphi}$. That is,*

$$V_{\mathit{eff}}(\bar{\varphi}) = \langle a|\mathcal{H}|a\rangle, \tag{3.7}$$

for $|a\rangle$ which obeys $\delta\langle a|\mathcal{H}|a\rangle = 0$ subject to the constraints $\langle a|a\rangle = 1$, $\langle a|\varphi|a\rangle = \bar{\varphi}$.

Proof. We first solve the analogous problem in quantum mechanics, then translate to field theory. In QM the problem is to find a state $|a\rangle$ such that $\langle a|H|a\rangle$ is stationary subject to $\langle a|a\rangle = 1$ and $\langle a|A|a\rangle = A_c$ for some self-adjoint operator A. Introducing Lagrange multipliers E and J for each constraint, the problem becomes

$$\delta\langle a|(H - E - JA)|a\rangle = 0 \quad \text{(unconstrained)} \tag{3.8}$$
$$\langle a|a\rangle = 1 \tag{3.9}$$
$$\langle a|A|a\rangle = A_c. \tag{3.10}$$

Since H is self-adjoint, this yields

$$(H - E - JA)|a\rangle = 0, \tag{3.11}$$

with solution $|a\rangle = |a(E, J)\rangle$. Using $\langle a|a\rangle = 1$, E is solved as a function of J, i.e. $E = E(J)$, and $\langle a|A|a\rangle = A_c$ gives $J = J(A_c)$. Thus

$$(H - JA)|a(J)\rangle = E(J)|a(J)\rangle, \tag{3.12}$$

where $|a(J)\rangle$ is a normalized eigenstate of $(H - JA)$. Taking the expectation value with $\langle a(J)|$ and differentiating with respect to J,

$$\begin{aligned} &\left(\frac{\delta}{\delta J}\langle a(J)|\right)\underbrace{(H - E(J) - JA)|a(J)\rangle}_{0} + \langle a(J)|\frac{\delta}{\delta J}(H - E(J) - JA)|a(J)\rangle \\ &\qquad + \langle a(J)|(H - E(J) - JA)\underbrace{\frac{\delta}{\delta J}|a(J)\rangle}_{0} = 0, \end{aligned} \tag{3.13}$$

which gives

$$-\frac{\delta E}{\delta J} - \underbrace{\langle a(J)|A|a(J)\rangle}_{A_c} = 0 \qquad \Rightarrow \qquad \frac{\delta E}{\delta J} = -A_c. \tag{3.14}$$

Substituting back into the expectation value of the eigenvalue equation,

$$\langle a(J)|H|a(J)\rangle = E(J) - J\frac{\delta E}{\delta J}, \tag{3.15}$$

which is the energy of the ground state of H in the space of states satisfying $\langle a(J)|A|a(J)\rangle = A_c$.

We now translate this to field theory. In field theory the only normalizable eigenstate of a Hamiltonian is the vacuum, so $|a(J)\rangle$ is identified with the vacuum of the sourced theory. Adding a source $J(x)\varphi(x)$ to the Hamiltonian density, adiabatically switched on at $t = 0$ and off at $t = T$ over spatial volume V, the adiabatic theorem gives

$$e^{iW(J)} = \langle 0^+|0^-\rangle = \exp\left[-iVT\,\varepsilon(J)\right], \tag{3.16}$$

so $-W(J)$ is the ground state action of the sourced theory. The Legendre transform structure,

$$\Gamma(\bar{\varphi}) = W(J) - J\frac{\delta W}{\delta J}, \qquad \frac{\delta W}{\delta J} = \bar{\varphi}, \tag{3.17}$$

is formally identical to the quantum-mechanical result above, identifying $-\Gamma(\bar{\varphi})$ as the ground state action of the sourceless theory in the sector with $\langle a|\varphi|a\rangle = \bar{\varphi}$. That is,

$$-\Gamma(\bar{\varphi}) = \langle a|H|a\rangle, \tag{3.18}$$

$$-\frac{\Gamma(\bar{\varphi})}{VT} = \frac{\langle a|H|a\rangle}{VT}, \tag{3.19}$$

and therefore

$$V_{\text{eff}}(\bar{\varphi}) = \langle a|\mathcal{H}|a\rangle \tag{3.20}$$

for the lowest-energy state satisfying $\langle a|a\rangle = 1$ and $\langle a|\varphi|a\rangle = \bar{\varphi}$. □

This result has an important consequence: since $V_{\text{eff}}(\bar{\varphi})$ is defined as a minimum of energy over a convex set of states, it is convex. Its minima therefore correspond to the stable equilibrium configurations of the quantum field theory, a fact that will be central to the analysis of phase transitions in later chapters.

3.3 Variational Definition of the Full Effective Action

The variational interpretation of V_{eff} established above can be generalized to the full effective action.

Theorem 2. $\Gamma(\varphi)$ *is the stationary value of*

$$\Gamma(\varphi) = \int_{-\infty}^{\infty} dt\,\langle\psi_-, t|\,i\partial_t - H\,|\psi_+, t\rangle \tag{3.21}$$

subject to the constraints

$$\langle\psi_-, t|\Phi(\mathbf{x})|\psi_+, t\rangle = \varphi(t, \mathbf{x}), \qquad \langle\psi_-, t|\psi_+, t\rangle = 1, \qquad \lim_{t\to\pm\infty} |\psi_\pm, t\rangle = |0\rangle. \tag{3.22}$$

Proof. The proof follows the same structure as Theorem 2. Introducing Lagrange multipliers $J(t, \mathbf{x})$ and $w(t)$ for the two constraints,

$$\Gamma_m(\varphi) = \int dt \, \langle\psi_-, t|i\partial_t - H|\psi_+, t\rangle + \int d^4x \, J(t, \mathbf{x}) \left(\langle\psi_-, t|\Phi|\psi_+, t\rangle - \varphi(t, \mathbf{x})\right) - w(t) \left(\langle\psi_-, t|\psi_+, t\rangle - 1\right). \quad (3.23)$$

Varying with respect to $\langle\psi_-, t|$, the last term drops out and since $\langle\delta\psi_-, t|$ is arbitrary,

$$\boxed{\left(i\partial_t - H + \int d^3x \, J(t, \mathbf{x}) \, \Phi(\mathbf{x})\right) |\psi_+, t\rangle = w(t)|\psi_+, t\rangle.} \quad (3.24)$$

Varying with respect to $|\psi_+, t\rangle$ and integrating by parts,

$$\int dt \, \langle\psi_-, t|i\partial_t|\delta\psi_+, t\rangle = [\langle\psi_-, t|\delta\psi_+, t\rangle]_{-\infty}^{\infty} - \int dt \, i \left(\partial_t\langle\psi_-, t|\right) |\delta\psi_+, t\rangle, \quad (3.25)$$

where the boundary term vanishes because $\lim_{t\to\pm\infty} |\psi_\pm, t\rangle = |0\rangle$. Hence

$$\int dt \, \langle\psi_-, t|i\partial_t|\delta\psi_+, t\rangle = -\int dt \, i \left(\partial_t\langle\psi_-, t|\right) |\delta\psi_+, t\rangle. \quad (3.26)$$

Since $|\delta\psi_+, t\rangle$ is arbitrary,

$$\delta_{|\psi_+,t\rangle}\Gamma_m(\varphi) = \int dt \, \langle\psi_-, t| \left(-i\overleftarrow{\partial_t} + H + \int d^3x \, J(\mathbf{x}) \, \Phi(\mathbf{x}) + w(t)\right) |\delta\psi_+, t\rangle = 0, \quad (3.27)$$

and taking the Hermitian conjugate,

$$\boxed{\left[i\partial_t - H + \int d^3x \, J(\mathbf{x}) \, \Phi(\mathbf{x})\right] |\psi_-, t\rangle = w^*(t)|\psi_-, t\rangle.} \quad (3.28)$$

Defining the rescaled states

$$|+, t\rangle = e^{i\int_{-\infty}^{t} dt' \, w(t')} \, |\psi_+, t\rangle, \quad (3.29)$$

$$|-, t\rangle = e^{-i\int_{t}^{\infty} dt' \, w^*(t')} \, |\psi_-, t\rangle, \quad (3.30)$$

these are asymptotic ground states satisfying the sourced Schrödinger equation with no inhomogeneous term. Their overlap gives

$$\langle-, t|+, t\rangle = e^{iW(J)} = \exp\left[i \int_{-\infty}^{\infty} dt \, w(t)\right], \quad (3.31)$$

from which

$$W(J) = \int_{-\infty}^{\infty} dt \, \langle\psi_-, t| \, i\partial_t - H + \int d^3x \, J \cdot \Phi \, |\psi_+, t\rangle. \quad (3.32)$$

As shown in Appendix D, $\delta W(J)/\delta J = \varphi$, so the Legendre transform gives

$$W(J) - \int d^4x \, J\varphi = \Gamma(\varphi) = \int_{-\infty}^{\infty} dt \, \langle\psi_-, t| \, i\partial_t - H \, |\psi_+, t\rangle, \quad (3.33)$$

which shows that $\Gamma(\varphi)$ is indeed the stationary value. $\square$

The equation (2.9) has a further physical interpretation. When the source vanishes, $\delta\Gamma/\delta\bar{\varphi} = 0$ becomes the exact quantum equation of motion for $\varphi(\mathbf{x}, t)$. Expanding Γ in powers of momentum about vanishing external momenta,

$$\Gamma(\varphi) = \int d^4x \left[-V_{\text{eff}}(\varphi) + \frac{1}{2} \left(\partial_\mu \varphi \, \partial^\mu \varphi \right) Z(\varphi) + \cdots \right], \tag{3.34}$$

and renormalizing the field so that $Z(\varphi) = 1$, the equation of motion becomes

$$\Box \, \varphi = -V'_{\text{eff}}(\varphi). \tag{3.35}$$

In the static limit this is precisely the classical Klein-Gordon equation, but with the classical potential replaced by the effective potential. Thus quantum corrections to the dynamics are fully captured by V_{eff}, which is yet another reason it is the central object of study.

Chapter 4

Toy Model: One-Loop Effective Potential and Renormalization

We illustrate the general formalism of the previous chapter with an explicit computation in the simplest setting: a single real scalar field with Lagrangian

$$\mathcal{L}(x) = \frac{1}{2}\,\partial_\mu\phi(x)\,\partial^\mu\phi(x) - V(\phi(x)), \tag{4.1}$$

and classical potential

$$V(\phi) = \frac{\alpha_0}{4!}\,\phi^4 - \frac{m_0^2}{2}\,\phi^2. \tag{4.2}$$

We work in units $\hbar = c = 1$ throughout.

4.1 The Classical Field and the Effective Action

The classical minimum of $V(\phi)$ sits at some ϕ_0, but the vacuum expectation value $\langle\phi\rangle$ of the quantum field need not coincide with it. It is defined by

$$\langle\phi\rangle \equiv \lim_{J\to 0} \frac{\langle 0^+|\phi_{\rm op}|0^-\rangle_J}{\langle 0^+|0^-\rangle_J}, \tag{4.3}$$

where the generating functional $W[J]$ encodes the full quantum theory via

$$\exp\frac{i}{\hbar}W[J] = \mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{S[\phi] + (J,\phi)\right\}, \tag{4.4}$$

with $S[\phi] = \int d^4x\,\mathcal{L}(x)$ and $(J,\phi) = \int d^4x\,J(x)\,\phi(x)$. The classical field, the VEV in the presence of an external source $J(x)$, is

$$\phi_c(x) \equiv \frac{\langle 0^+|\phi_{\rm op}(x)|0^-\rangle_J}{\langle 0^+|0^-\rangle_J} = \frac{\delta W[J]}{\delta J(x)}, \tag{4.5}$$

and $\langle\phi\rangle$ is recovered as $\lim_{J\to 0}\phi_c(x)$. The loop expansion is an expansion in powers of $\hbar\,c\,\alpha$, but it performs an infinite resummation of Feynman graphs rather than a perturbative truncation.

The natural question is: given a desired profile ϕ_c, what source J produces it? To answer this, we trade J for ϕ_c as the independent variable via the Legendre transform,

$$\Gamma[\phi_c] \equiv W[J] - (J,\phi_c), \tag{4.6}$$

where J is eliminated in favour of ϕ_c by inverting (4.5). The functional $\Gamma[\phi_c]$ is the effective action, so named because it is a functional of the classical field ϕ_c and thus plays the same role for the quantum theory as $S[\phi]$ does classically. Differentiating (4.6) and using (4.5),

$$\begin{aligned}\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} &= \frac{\delta W[J]}{\delta\phi_c(x)} - J(x) - \int d^4y\,\frac{\delta J(y)}{\delta\phi_c(x)}\,\phi_c(y),\\ \frac{\delta W[J]}{\delta\phi_c(x)} &= \int d^4y\,\frac{\delta W[J]}{\delta J(y)}\frac{\delta J(y)}{\delta\phi_c(x)} = \int d^4y\,\frac{\delta J(y)}{\delta\phi_c(x)}\,\phi_c(y),\end{aligned} \tag{4.7}$$

so that

$$\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} = -J(x). \tag{4.8}$$

At $J=0$, translational invariance forces ϕ_c to be a constant, so $\langle\phi\rangle$ is the root of

$$\left.\frac{d\Gamma[\phi_c]}{d\phi_c}\right|_{\phi_c=\langle\phi\rangle} = 0. \tag{4.9}$$

4.2 The Effective Action as a Generator of 1PI Green's Functions

Expanding $\Gamma[\phi_c]$ in powers of ϕ_c,

$$\Gamma[\phi_c] = \sum_{n=0}^{\infty}\frac{1}{n!}\int d^4x_1\cdots d^4x_n\,\Gamma_n(x_1,\ldots,x_n)\,\phi_c(x_1)\cdots\phi_c(x_n), \tag{4.10}$$

we claim that Γ_n is the n-point 1PI Green function, the sum of all connected 1PI Feynman graphs with n external legs. To see this, consider the functional integral

$$\mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{\Gamma[\phi]+(J,\phi)\right\}. \tag{4.11}$$

In the limit $\hbar\to 0$ the integrand is dominated by its saddle point, which occurs at $\phi=\phi_c$ since (4.8) is precisely the saddle-point condition. Thus

$$\mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{\Gamma[\phi]+(J,\phi)\right\} \xrightarrow{\hbar\to 0} \mathcal{N}\exp\frac{i}{\hbar}\left\{\Gamma[\phi_c]+(J,\phi_c)\right\}, \tag{4.12}$$

and by (4.6) the right-hand side equals $\exp\frac{i}{\hbar}W[J]$, so

$$\exp\frac{i}{\hbar}W[J] \xrightarrow{\hbar\to 0} \mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{\Gamma[\phi]+(J,\phi)\right\}. \tag{4.13}$$

In this limit the right-hand side is the sum of all tree graphs of a theory whose action is $\Gamma[\phi]$, with vertices given by the Γ_n of (4.10). Since $W[J]$ generates all connected Green functions of the original theory, and any connected graph decomposes into 1PI components, it follows that Γ_n is a 1PI Green function of the original theory. $\square$

4.3 The Effective Potential

Alternatively, $\Gamma[\phi_c]$ may be expanded in derivatives of ϕ_c,

$$\Gamma[\phi_c] = \int d^4x \left[-U(\phi_c(x)) + \frac{1}{2}\,\partial_\mu\phi_c(x)\,\partial^\mu\phi_c(x)\,Z(\phi_c(x)) + \cdots \right], \tag{4.14}$$

where $U(\phi_c)$ is the effective potential. For a constant configuration $\phi_c(x) = \rho$ this reduces to

$$\Gamma[\rho] = -\Omega\,U(\rho), \tag{4.15}$$

where Ω is the total spacetime volume. To connect $U(\rho)$ to the 1PI Green functions, we pass to momentum space. The Fourier transforms of ϕ_c and Γ_n are defined by

$$\tilde{\phi}_c(k) = \int d^4x\, e^{ik\cdot x}\,\phi_c(x), \tag{4.16}$$

$$\phi_c(x) = \int \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot x}\,\tilde{\phi}_c(k), \tag{4.17}$$

$$\tilde{\Gamma}_n(k_1,\ldots,k_n)\,(2\pi)^4\,\delta^4(k_1+\cdots+k_n) = \int d^4x_1\cdots d^4x_n\,\Gamma_n(x_1,\ldots,x_n)\,e^{i(k_1x_1+\cdots+k_nx_n)}. \tag{4.18}$$

Substituting these into (4.10),

$$\begin{aligned}
\Gamma[\phi_c] &= \sum_{n=0}^{\infty}\frac{1}{n!}\int d^4x_1\cdots d^4x_n\,\Gamma_n(x_1,\ldots,x_n)\,\phi_c(x_1)\cdots\phi_c(x_n)\\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\int\frac{d^4k_1}{(2\pi)^4}\cdots\frac{d^4k_n}{(2\pi)^4}\int d^4x_1\cdots d^4x_n\,\Gamma_n(x_1,\ldots,x_n)\,e^{-i(k_1x_1+\cdots+k_nx_n)}\,\tilde{\phi}(k_1)\cdots\tilde{\phi}(k_n)\\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\int\frac{d^4k_1}{(2\pi)^4}\cdots\frac{d^4k_n}{(2\pi)^4}\,(2\pi)^4\,\delta^4(k_1+\cdots+k_n)\,\tilde{\Gamma}_n(k_1,\ldots,k_n)\,\tilde{\phi}(k_1)\cdots\tilde{\phi}(k_n).
\end{aligned} \tag{4.19}$$

Setting $\phi_c(x) = \rho$ gives $\tilde{\phi}_c(k) = (2\pi)^4\delta^4(k)\,\rho$, so each momentum integral collapses,

$$\begin{aligned}
\Gamma[\rho] = \sum_{n=0}^{\infty}\frac{1}{n!}\int\frac{d^4k_1}{(2\pi)^4}\cdots\frac{d^4k_n}{(2\pi)^4}\,&(2\pi)^4\,\delta^4(k_1+\cdots+k_n)\,\tilde{\Gamma}_n(k_1,\ldots,k_n)\\
&\cdot\,(2\pi)^4\delta^4(k_1)\,\rho\cdots(2\pi)^4\delta^4(k_n)\,\rho\\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\underbrace{(2\pi)^4\delta^4(0)}_{\Omega}\,\rho^n\,\tilde{\Gamma}_n(0,\ldots,0),
\end{aligned} \tag{4.20}$$

and therefore

$$\Gamma[\rho] = \Omega\sum_{n=0}^{\infty}\frac{\rho^n}{n!}\,\tilde{\Gamma}_n(0), \tag{4.21}$$

where $\tilde{\Gamma}_n(0) \equiv \tilde{\Gamma}_n(0,\ldots,0)$ and $\Omega = (2\pi)^4\delta^4(0)$ is the total spacetime volume. Comparing (4.21) with $\Gamma[\rho] = -\Omega\,U(\rho)$,

$$U(\rho) = -\sum_{n=0}^{\infty}\frac{\rho^n}{n!}\,\tilde{\Gamma}_n(0), \tag{4.22}$$

which is simply the statement that the Taylor coefficients of U are the zero-momentum 1PI Green functions,

$$-\frac{d^n U(\rho)}{d\rho^n}\bigg|_{\rho=0} = \tilde{\Gamma}_n(0). \tag{4.23}$$

The effective potential is thus entirely determined by the 1PI Green functions of the original theory evaluated at vanishing external momenta. In the next section we compute it explicitly at one loop.

4.4 Spontaneous Symmetry Breaking and the Shifted Field

When spontaneous symmetry breaking occurs, it is more natural to work with the shifted field

$$\eta(x) = \phi(x) - \langle\phi\rangle, \tag{4.24}$$

so that the effective potential in terms of the shifted variable becomes

$$U(\rho - \langle\phi\rangle) = -\sum_{n=0}^{\infty} \frac{[\rho - \langle\phi\rangle]^n}{n!}\, \widetilde{\Gamma}_n^{(s)}(0), \tag{4.25}$$

where $\widetilde{\Gamma}_n^{(s)}(0)$ are the 1PI Green functions defined with respect to $\eta(x)$. The physically relevant quantities extracted from $U(\rho)$ at the minimum $\rho = \langle\phi\rangle$ are

$$\frac{dU(\rho)}{d\rho}\bigg|_{\rho=\langle\phi\rangle} = 0, \tag{4.26}$$

which determines $\langle\phi\rangle$, together with the renormalized mass and coupling constant,

$$\frac{d^2U(\rho)}{d\rho^2}\bigg|_{\rho=\langle\phi\rangle} = m^2, \qquad \frac{d^4U(\rho)}{d\rho^4}\bigg|_{\rho=\langle\phi\rangle} = \alpha. \tag{4.27}$$

Since $m^2 \geq 0$, these conditions mean that $\langle\phi\rangle$ sits at a minimum of $U(\rho)$, in direct analogy with ρ_0 being a minimum of the classical potential $V(\rho)$. The parameters m^2 and α are the free parameters of the renormalized theory.

4.5 One-Loop Effective Potential

We now compute $U(\rho)$ to one-loop order via saddle-point integration. Starting from

$$\exp\frac{i}{\hbar}W[J] = \mathcal{N}\int \mathcal{D}\phi\, \exp\frac{i}{\hbar}\left\{S[\phi] + (J,\phi)\right\}, \tag{4.28}$$

the saddle point occurs at $\phi = \phi_0$, defined by

$$\frac{\delta S[\phi]}{\delta\phi(x)}\bigg|_{\phi=\phi_0} = -J(x). \tag{4.29}$$

Here $\phi_0(x)$ is a functional of J, and as $J \to 0$ it satisfies the classical equation of motion. Expanding $S[\phi]$ about ϕ_0,

$$\begin{aligned} S[\phi_0 + \phi_c] = S[\phi_0] + \int d^4x \; \phi_c(x) \left[\frac{\delta S[\phi]}{\delta \phi}\right]_{\phi=\phi_0} \\ + \frac{1}{2}\int d^4x \, d^4y \; \phi_c(x)\, \phi_c(y) \left[\frac{\delta^2 S[\phi]}{\delta\phi(x)\,\delta\phi(y)}\right]_{\phi=\phi_0} \\ + S_2[\phi_c, \phi_0], \end{aligned} \tag{4.30}$$

where S_2 collects all higher-order terms. Introducing the propagator function

$$\langle x|\, i\Delta^{-1}[\phi_0]\,|y\rangle \equiv \left.\frac{\delta^2 S[\phi]}{\delta\phi(x)\,\delta\phi(y)}\right|_{\phi=\phi_0}, \tag{4.31}$$

the expansion becomes

$$S[\phi_0 + \phi_c] = S[\phi_0] - (J, \phi_c) + \frac{1}{2}\left(\phi_c,\, i\Delta^{-1}[\phi_0]\,\phi_c\right) + S_2[\phi_c, \phi_0]. \tag{4.32}$$

Substituting into the path integral and shifting $\phi \to \phi + \phi_0$,

$$\begin{aligned} \exp\frac{i}{\hbar}W[J] &= \mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{S[\phi+\phi_0] + (\phi+\phi_0, J)\right\} \\ &= \mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{S[\phi_0] + \frac{1}{2}(\phi,\, i\Delta^{-1}[\phi_0]\,\phi) + S_2[\phi,\phi_0] + (\phi_0, J)\right\}, \end{aligned} \tag{4.33}$$

which factors as

$$= \;\exp\frac{i}{\hbar}\left\{S[\phi_0] + (\phi_0, J)\right\}\cdot\left\{\mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{2\hbar}(\phi,\, i\Delta^{-1}[\phi_0]\,\phi)\right\}\cdot\left\langle \exp\frac{i}{\hbar}S_2\right\rangle, \tag{4.34}$$

where the Gaussian integral evaluates to $\left\{\det i\Delta^{-1}[\phi_0]\right\}^{-1/2}$ and $\langle\,\cdot\,\rangle$ denotes the functional average with respect to the Gaussian weight $\exp\frac{i}{2\hbar}(\phi,\, i\Delta^{-1}[\phi_0]\,\phi)$. Taking the logarithm,

$$W[J] = S[\phi_0] + (\phi_0, J) + \frac{i\hbar}{2}\ln\det\left\{i\Delta^{-1}[\phi_0]\right\} + W_2[J], \tag{4.35}$$

where $W_2[J] = -i\hbar\ln\langle\exp\frac{i}{\hbar}S_2\rangle$.

To determine the order of $W_2[J]$, rescale $\phi = \hbar^{1/2}\tilde{\phi}$,

$$\begin{aligned} \hbar^{-1}S_2[\phi,\phi_0] &= \hbar^{-1}\phi\phi\phi\left[\frac{\delta^3 S}{\delta\phi^3}\right]_{\phi=\phi_0} + \cdots \\ &= \hbar^{-1/2}\tilde{\phi}\tilde{\phi}\tilde{\phi}\left[\frac{\delta^3 S}{\delta\phi^3}\right]_{\phi=\phi_0} + \cdots, \end{aligned} \tag{4.36}$$

so that

$$W_2[J] = -i\hbar\ln\left[1 + \mathcal{O}(\hbar^{1/2})\right] \sim \mathcal{O}(\hbar^{3/2}). \tag{4.37}$$

Since the loop expansion of $W[J]$ contains only integer powers of $\hbar$, the fractional contribution must actually vanish and $W_2[J] \sim \mathcal{O}(\hbar^2)$. Thus the first two terms in (4.35) are the tree-level approximation and the third term is the complete one-loop correction,

$$W[J] = S[\phi_0] + (\phi_0, J) + \frac{i\hbar}{2}\ln\det\left\{i\Delta^{-1}[\phi_0]\right\} + \mathcal{O}(\hbar^2). \tag{4.38}$$

To obtain $\Gamma[\phi_c] = W[J] - (J, \phi_c)$, we express $S[\phi_0]$ as a functional of ϕ_c. At tree level $\phi_c = \phi_0$, so $\phi_c - \phi_0 \equiv \phi_1 \sim \mathcal{O}(\hbar)$,

$$\phi_c(x) \equiv \frac{\delta W[J]}{\delta J(x)} = \phi_0(x) + \phi_1(x). \tag{4.39}$$

Expanding $S[\phi_0] = S[\phi_c - \phi_1]$ about ϕ_c,

$$\begin{aligned} S[\phi_0] &= S[\phi_c] - \int d^4x\, \phi_1(x) \left\{ \frac{\delta S[\phi_0]}{\delta \phi} \right\}_{\phi=\phi_c} + \mathcal{O}(\hbar^2) \\ &= S[\phi_c] + (\phi_1, J) + \mathcal{O}(\hbar^2). \end{aligned} \tag{4.40}$$

Substituting into $\Gamma[\phi_c]$,

$$\begin{aligned} \Gamma[\phi_c] &= S[\phi_c] + \underbrace{(\phi_1, J)}_{\mathcal{O}(\hbar)} + (\phi_c, J) + \frac{i\hbar}{2} \ln\det\left\{ i\Delta^{-1}[\phi_c] \right\} + \mathcal{O}(\hbar^2) \\ &= S[\phi_c] + (\phi_c, J) + \frac{i\hbar}{2} \ln\det\left\{ i\Delta^{-1}[\phi_c] \right\} + \mathcal{O}(\hbar^2), \end{aligned} \tag{4.41}$$

where the (ϕ_1, J) term is absorbed into $\mathcal{O}(\hbar^2)$. Setting $J = 0$ and $\phi_c(x) = \rho$ (const), and using $\Gamma[\rho] = -\Omega\, U(\rho)$ and $S[\rho] = -\Omega\, V(\rho)$,

$$-\Omega\, U(\rho) = -\Omega\, V(\rho) + \frac{i\hbar}{2} \ln\det\left\{ i\Delta^{-1}[\rho] \right\} + \mathcal{O}(\hbar^2), \tag{4.42}$$

$$\boxed{U(\rho) = V(\rho) - \frac{i\hbar}{2\Omega} \ln\det\left\{ i\Delta^{-1}[\rho] \right\} + \mathcal{O}(\hbar^2).} \tag{4.43}$$

The determinant is evaluated using $\ln\det = \operatorname{Tr}\ln$,

$$\begin{aligned} \ln\det\left\{ i\Delta^{-1}[\rho] \right\} &= \operatorname{Tr}\ln\left\{ i\Delta^{-1}[\rho] \right\} \\ &= \int \frac{d^4k}{(2\pi)^4} \ln\langle k|\, i\Delta^{-1}[\rho]\, |k\rangle, \end{aligned} \tag{4.44}$$

so that

$$U(\rho) = V(\rho) - \frac{i\hbar}{2} \int \frac{d^4k}{(2\pi)^4}\, \Omega^{-1} \ln\langle k|\, i\Delta^{-1}[\rho]\, |k\rangle + \mathcal{O}(\hbar^2). \tag{4.45}$$

To evaluate $\Omega^{-1}\langle k|\, i\Delta^{-1}[\rho]\, |k\rangle$, expand $\phi(x) = \rho + \phi_1(x)$ with ρ constant and $\partial V/\partial\phi|_{\phi=\rho} = 0$,

$$\begin{aligned} S[\phi] &= \int d^4x \left[\frac{1}{2}\, \partial_\mu \phi_1\, \partial^\mu \phi_1 - V(\phi_1 + \rho) \right] \\ &= -\frac{1}{2} \int d^4x\; \phi_1 \left[\Box + V''(\rho) \right] \phi_1 + \cdots . \end{aligned} \tag{4.46}$$

By (4.31),

$$\begin{aligned} \langle x|\, i\Delta^{-1}[\rho]\, |y\rangle &= \left. \frac{\delta^2 S[\phi]}{\delta \phi_1(x)\, \delta \phi_1(y)} \right|_{\phi_1=0} \\ &= -\left[\Box + V''(\rho) \right] \delta^4(x-y). \end{aligned} \tag{4.47}$$

Taking the matrix element between momentum eigenstates (see Appendix E),

$$\Omega^{-1}\langle k|\, i\Delta^{-1}[\rho]\,|k\rangle = k^2 - V''(\rho). \tag{4.48}$$

Substituting into (4.45) and rotating to Euclidean momentum space via $k^0 \to ik_E^0$, $k^2 \to -k_E^2$, $d^4k = i\,d^4k_E$,

$$U(\rho) = V(\rho) + \frac{\hbar}{2}\int \frac{d^4k_E}{(2\pi)^4}\ln\left[k_E^2 + V''(\rho)\right] + \mathcal{O}(\hbar^2), \tag{4.49}$$

where the $i\pi$ contribution from $\ln(-k_E^2 - V'')$ is a field-independent constant and has been dropped. The momentum integral is evaluated in Appendix F, with the result

$$\frac{\hbar}{2}\int \frac{d^4k_E}{(2\pi)^4}\ln(k_E^2 + V'') \;=\; \hbar\left[\frac{\Lambda^2}{32\pi^2}V''(\rho) + \frac{[V''(\rho)]^2}{64\pi^2}\left(-\frac{1}{2} + \ln\frac{V''(\rho)}{\Lambda^2}\right) + \mathcal{O}\left(\frac{1}{\Lambda^2}\right)\right]. \tag{4.50}$$

We can therefore write

$$U(\rho) = V(\rho) + \hbar\, V_1(\rho) + \mathcal{O}(\hbar^2), \tag{4.51}$$

where the one-loop correction is

$$V_1(\rho) = \frac{\Lambda^2}{32\pi^2}V''(\rho) + \frac{[V''(\rho)]^2}{64\pi^2}\left[-\frac{1}{2} + \ln\frac{V''(\rho)}{\Lambda^2}\right]. \tag{4.52}$$

To separate the divergent from the convergent part, introduce an arbitrary renormalization scale μ via

$$\ln\frac{V''}{\Lambda^2} = \ln\frac{V''}{\mu^2} + \ln\frac{\mu^2}{\Lambda^2}, \tag{4.53}$$

so that

$$\begin{aligned} V_1(\rho) &= \frac{\Lambda^2}{32\pi^2}V''(\rho) + \frac{(V''(\rho))^2}{64\pi^2}\left[-\frac{1}{2} + \ln\frac{V''}{\mu^2} + \ln\frac{\mu^2}{\Lambda^2}\right] \\ &= \underbrace{\frac{1}{32\pi^2}\left[\Lambda^2 V''(\rho) - \frac{1}{2}[V''(\rho)]^2\ln\frac{\Lambda^2}{\mu^2}\right]}_{\text{divergent}} + \underbrace{\frac{(V'')^2}{64\pi^2}\left[-\frac{1}{2} + \ln\frac{V''}{\mu^2}\right]}_{\text{convergent}}. \end{aligned} \tag{4.54}$$

The divergent part must be removed by renormalization, which we turn to next.

4.6 Renormalization

The theory is renormalizable if the counterterms needed to cancel the divergences are of the same form as terms already present in the Lagrangian. For a classical potential $V(\rho)$ that is a polynomial of degree n, the second derivative $V''(\rho)$ is of degree $n-2$, and the one-loop correction $V_1(\rho)$ in (4.52) is of degree $2(n-2) = 2n-4$ from the $(V'')^2$ term. Renormalizability then requires

$$2n - 4 \le n \quad \Rightarrow \quad n \le 4, \tag{4.55}$$

a condition satisfied by our choice of $V(\rho)$.

The bare parameters are split into renormalized parameters and counterterms,

$$\alpha_0 = \alpha_1 + \delta\alpha, \tag{4.56}$$

$$m_0^2 = m_1^2 + \delta m^2, \tag{4.57}$$

where $\delta\alpha$ and δm^2 are divergent in perturbation theory but α_1 and m_1^2 are required to be finite. Since the counterterms only arise at the quantum level, $\delta\alpha$, $\delta m^2 \sim \mathcal{O}(\hbar)$. The Lagrangian becomes

$$\mathcal{L}(x) = \frac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi - V(\phi) + \underbrace{\frac{\delta m^2}{2}\phi^2 - \frac{\delta\alpha}{4!}\phi^4}_{\text{counterterms}}, \tag{4.58}$$

with the renormalized classical potential

$$V(\phi) = \frac{\alpha_1}{4!}\phi^4 - \frac{m_1^2}{4}\phi^2. \tag{4.59}$$

The classical vacuum value ρ_0 is found from

$$\frac{dV(\phi)}{d\phi} = \frac{\alpha_1}{3!}\phi^3 - \frac{m_1^2}{2}\phi = 0$$

$$\Rightarrow \phi\left(\frac{\alpha_1}{3}\phi^2 - m_1^2\right) = 0, \tag{4.60}$$

giving $\phi = 0$ or

$$\rho_0 = m_1\left(\frac{3}{\alpha_1}\right)^{\frac{1}{2}}. \tag{4.61}$$

Since the counterterms are of order $\hbar$, they contribute at the same order as the one-loop correction and the full effective potential is

$$U(\rho) = V(\rho) + \hbar\,V_1(\rho) + \frac{\delta\alpha}{4!}\rho^4 - \frac{\delta m^2}{2}\rho^2. \tag{4.62}$$

Choosing $\delta\alpha$ and δm^2 to precisely cancel the divergent part of $V_1(\rho)$,

$$\delta\alpha = \frac{3\alpha_1^2\hbar}{32\pi^2}\ln\frac{\Lambda^2}{\mu^2}, \tag{4.63}$$

$$\delta m^2 = \frac{\alpha_1\hbar}{32\pi^2}\left(\Lambda^2 + \frac{m_1^2}{2}\ln\frac{\Lambda^2}{\mu^2}\right), \tag{4.64}$$

where the right-hand sides are ambiguous up to finite additive terms, the effective potential becomes

$$U(\rho) = V(\rho) + \frac{\hbar}{64\pi^2}\left[V''(\rho)\right]^2\left[-\frac{1}{2} + \ln\frac{V''(\rho)}{\mu^2}\right]. \tag{4.65}$$

4.7 Physical Parameters from the Effective Potential

We now extract the VEV, the renormalized mass, and the renormalized coupling constant from (4.65). Setting $k \equiv \hbar/(32\pi^2)$ for brevity, the successive derivatives of $U(\rho)$ are computed using $V = \frac{\alpha_1}{4!}\rho^4 - \frac{m_1^2}{4}\rho^2$, so $V' = \frac{\alpha_1}{3!}\rho^3 - \frac{m_1^2}{2}\rho$, $V'' = \frac{\alpha_1}{2}\rho^2 - \frac{m_1^2}{2}$, $V''' = \alpha_1\rho$, $V'''' = \alpha_1$, $V''''' = 0$.

Differentiating (4.65),

$$U' = V' + \frac{\hbar}{64\pi^2}\left[2V''V'''\left(-\frac{1}{2} + \ln\frac{V''}{\mu^2}\right) + (V'')^2\cdot\frac{1}{V''}\cdot\frac{V'''}{\mu^2}\cdot\mu^2\right]$$

$$= V' + \frac{\hbar}{64\pi^2}\left[-V''V''' + 2V''V'''\ln\frac{V''}{\mu^2} + V''V'''\right], \quad (4.66)$$

$$\Rightarrow U' = V' + k\,V''V'''\ln\frac{V''}{\mu^2}. \quad (4.67)$$

Differentiating again,

$$U'' = V'' + k\left[(V''')^2\ln\frac{V''}{\mu^2} + V''V''''\ln\frac{V''}{\mu^2} + V''V'''\cdot\frac{1}{V''}\cdot\frac{V'''}{\mu^2}\cdot\mu^2\right], \quad (4.68)$$

$$\Rightarrow U'' = V'' + k\left[\left\{(V''')^2 + \alpha_1 V''\right\}\ln\frac{V''}{\mu^2} + (V''')^2\right]. \quad (4.69)$$

The third derivative,

$$U''' = V''' + k\Bigg[\left\{(V''')^2 + \alpha_1 V''\right\}'\ln\frac{V''}{\mu^2} + \left\{(V''')^2 + \alpha_1 V''\right\}\left(\ln\frac{V''}{\mu^2}\right)' + 2V'''V''''\Bigg]$$

$$= V''' + k\left[(2V'''V'''' + \alpha_1 V''')\ln\frac{V''}{\mu^2} + \left\{(V''')^2 + \alpha_1 V''\right\}\frac{V'''}{V''} + 2\alpha_1 V'''\right], \quad (4.70)$$

$$\Rightarrow U''' = V''' + k\left[3\alpha_1 V'''\ln\frac{V''}{\mu^2} + \frac{(V''')^3}{V''} + 3\alpha_1 V'''\right]. \quad (4.71)$$

The fourth derivative,

$$U'''' = V'''' + k\Bigg[3\alpha_1 V''''\ln\frac{V''}{\mu^2} + 3\alpha_1 V'''\cdot\frac{1}{V''}\cdot\frac{V'''}{\mu^2}\cdot\mu^2 + \frac{3(V''')^2V''''}{V''} - \frac{(V''')^3}{(V'')^2}\cdot V''' + 3\alpha_1 V''''\Bigg]$$

$$= \alpha_1 + k\left[3\alpha_1^2\ln\frac{V''}{\mu^2} + 3\alpha_1\frac{(V''')^2}{V''} + \frac{3\alpha_1(V''')^2}{V''} - \frac{(V''')^4}{(V'')^2} + 3\alpha_1^2\right]$$

$$= \alpha_1 + k\left[3\alpha_1^2\ln\frac{V''}{\mu^2} + \frac{6\alpha_1(V''')^2}{V''} - \frac{(V''')^4}{(V'')^2} + 3\alpha_1^2\right], \quad (4.72)$$

$$\Rightarrow U'''' = \alpha_1 + k\alpha_1^2\left[3\ln\frac{V''}{\mu^2} + \frac{6\alpha_1\rho^2}{V''} - \frac{\alpha_1^2\rho^4}{(V'')^2} + 3\right]. \quad (4.73)$$

The VEV

The minimum $\langle\phi\rangle = \rho_0 + \rho_1$ is found by expanding $U'(\rho_0 + \rho_1) = 0$ to first order,

$$U'(\rho_0) + \rho_1 U''(\rho_0) = 0 \quad \Rightarrow \quad \rho_1 = -\frac{U'(\rho_0)}{U''(\rho_0)}. \quad (4.74)$$

We first evaluate $V''(\rho_0)$ and $V'''(\rho_0)$ at the classical minimum $\rho_0^2 = 3m_1^2/\alpha_1$,

$$V''(\rho_0) = \frac{\alpha_1}{2}\rho_0^2 - \frac{m_1^2}{2} = \frac{\alpha_1}{2}\cdot\frac{3m_1^2}{\alpha_1} - \frac{m_1^2}{2} = \frac{3m_1^2}{2} - \frac{m_1^2}{2} = m_1^2, \tag{4.75}$$

$$\Rightarrow V''(\rho_0) = m_1^2. \tag{4.76}$$

Also, $V'(\rho_0) = 0$ by definition of ρ_0, so from (4.67),

$$U'(\rho_0) = k\, m_1^2\, \alpha_1\, \rho_0 \ln\frac{m_1^2}{\mu^2}. \tag{4.77}$$

For $U''(\rho_0)$, the $\mathcal{O}(\alpha_1^2)$ terms are dropped,

$$\begin{aligned} U''(\rho_0) &= m_1^2 + k\left[(\alpha_1^2\rho_0^2 + \alpha_1 m_1^2)\ln\frac{m_1^2}{\mu^2} + \alpha_1^2\rho_0^2\right] \\ &= m_1^2 + k\,\alpha_1 m_1^2 \ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2), \end{aligned} \tag{4.78}$$

$$\Rightarrow U''(\rho_0) = m_1^2\left(1 + \alpha_1 k \ln\frac{m_1^2}{\mu^2}\right) + \mathcal{O}(\alpha_1^2). \tag{4.79}$$

Hence,

$$\begin{aligned} \rho_1 &= -\frac{k\, m_1^2\, \alpha_1\, \rho_0 \ln\frac{m_1^2}{\mu^2}}{m_1^2\left(1 + \alpha_1 k \ln\frac{m_1^2}{\mu^2}\right)} \\ &= -k\,\alpha_1\,\rho_0 \ln\frac{m_1^2}{\mu^2}\left(1 + \alpha_1 k \ln\frac{m_1^2}{\mu^2}\right)^{-1} \\ &= -k\alpha_1\rho_0 \ln\frac{m_1^2}{\mu^2}\left[1 - \alpha_1 k \ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2)\right] \\ &= -k\alpha_1\rho_0 \ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2), \end{aligned} \tag{4.80}$$

$$\frac{\rho_1}{\rho_0} = -\frac{\hbar\alpha_1}{32\pi^2}\ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2). \tag{4.81}$$

Setting $\hbar = 1$, the full VEV is

$$\boxed{\langle\phi\rangle = m_1\left(\frac{3}{\alpha_1}\right)^{\frac{1}{2}}\left[1 - \frac{\alpha_1}{32\pi^2}\ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2)\right].} \tag{4.82}$$

The Renormalized Mass

The renormalized mass is $m^2 \equiv U''(\rho_0 + \rho_1)$. Expanding,

$$m^2 = U''(\rho_0) + \rho_1\, U'''(\rho_0). \tag{I}$$

From (4.71), dropping $\mathcal{O}(\alpha_1^2)$ terms,

$$\begin{aligned} U'''(\rho_0) &= \alpha_1\rho_0 + k\left[3\alpha_1^2\rho_0 \ln\frac{m_1^2}{\mu^2} + \frac{(\alpha_1\rho_0)^3}{m_1^2} + 3\alpha_1^2\rho_0\right] \\ &= \alpha_1\rho_0 + \mathcal{O}(\alpha_1^2), \end{aligned} \tag{II}$$

$$U'''(\rho_0) = \alpha_1\rho_0 + k\left[3\alpha_1^2\rho_0 \ln\frac{m_1^2}{\mu^2}\right]. \tag{III}$$

Also,

$$U''(\rho_0) = m_1^2 + k\left[(\alpha_1^2\rho_0^2 + \alpha_1 m_1^2)\ln\frac{m_1^2}{\mu^2} + \alpha_1^2\rho_0^2\right]. \tag{4.83}$$

Substituting (II) and (III) into (I), using $\rho_1 = -k\alpha_1\rho_0 \ln\frac{m_1^2}{\mu^2}$ and $\rho_1 U'''(\rho_0) = -k\alpha_1\rho_0\ln\frac{m_1^2}{\mu^2}\cdot 3m_1^2 + \mathcal{O}(\alpha_1^2)$,

$$\begin{aligned}
m^2 &= m_1^2\left(1 + 4\alpha_1 k\ln\frac{m_1^2}{\mu^2} + 3\alpha_1 k\right) - 3m_1^2\alpha_1 k\ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2) \\
&= m_1^2\left[1 + 4\alpha_1 k\ln\frac{m_1^2}{\mu^2} + 3\alpha_1 k - 3\alpha_1 k\ln\frac{m_1^2}{\mu^2}\right] + \mathcal{O}(\alpha_1^2) \\
&= m_1^2\left[1 + \alpha_1 k\ln\frac{m_1^2}{\mu^2} + 3\alpha_1 k\right] + \mathcal{O}(\alpha_1^2) \\
&= m_1^2\left[1 + 3\alpha_1 k\left(1 + \frac{1}{3}\ln\frac{m_1^2}{\mu^2}\right)\right] + \mathcal{O}(\alpha_1^2),
\end{aligned} \tag{4.84}$$

$$\Rightarrow \quad \boxed{m^2 = m_1^2\left[1 + \frac{3\alpha_1}{32\pi^2}\left(1 + \frac{1}{3}\ln\frac{m_1^2}{\mu^2}\right) + \mathcal{O}(\alpha_1^2)\right].} \tag{4.85}$$

The Renormalized Coupling Constant

The renormalized coupling is $\alpha \equiv U''''(\rho_0 + \rho_1)$. Evaluating (4.73) at $\rho = \rho_0$ and keeping terms to $\mathcal{O}(\alpha_1)$,

$$\begin{aligned}
\alpha &= \alpha_1 + k\alpha_1^2\left[3\ln\frac{V''}{\mu^2} + \frac{6\alpha_1\rho^2}{V''} - \frac{\alpha_1^2\rho^4}{(V'')^2} + 3\right] \\
&= \alpha_1\left[1 + 3\alpha_1 k\left(1 + \ln\frac{V''}{\mu^2}\right) + \mathcal{O}(\alpha_1^2)\right],
\end{aligned} \tag{4.86}$$

$$\Rightarrow \quad \boxed{\alpha = \alpha_1\left[1 + \frac{3\alpha_1}{32\pi^2}\left(1 + \ln\frac{m_1^2}{\mu^2}\right) + \mathcal{O}(\alpha_1^2)\right].} \tag{4.87}$$

4.8 The Massless Case and the Running Coupling Constant

We now consider the massless limit $m_1 = 0$, in which

$$V' = \frac{\alpha_1}{6}\rho^3, \qquad\qquad V'' = \frac{\alpha_1\rho^2}{2}. \tag{4.88}$$

From (4.73),

$$\begin{aligned}
U''''(\rho) &= \alpha_1 + k\alpha_1^2\left[3\ln\frac{\alpha_1\rho^2}{2\mu^2} + \frac{6\alpha_1\rho^2}{\frac{\alpha_1\rho^2}{2}} - \frac{\alpha_1^2\rho^4}{\frac{\alpha_1^2\rho^4}{4}} + 3\right] \\
&= \alpha_1 + k\alpha_1^2\left[3\ln\frac{\alpha_1\rho^2}{2\mu^2} + 12 - 4 + 3\right],
\end{aligned} \tag{4.89}$$

$$\Rightarrow U''''(\rho) = \alpha_1 + k\alpha_1^2 \left[11 + 3\ln\frac{\alpha_1\rho^2}{2\mu^2} \right]. \tag{4.90}$$

Setting $\alpha_1 \equiv \alpha$ and defining the running coupling constant $\alpha(\rho) \equiv U''''(\rho)$,

$$\alpha(\rho) = \alpha + \frac{k\alpha^2}{1} \left[11 + 3\ln\frac{\alpha\rho^2}{2\mu^2} \right]. \tag{4.91}$$

We introduce a new scale μ_0 defined by the renormalization condition $U''''(\mu_0) = \alpha$, which gives

$$\begin{aligned} \alpha &= \alpha + k\alpha^2 \left[11 + 3\ln\frac{\alpha\mu_0^2}{2\mu^2} \right] \\ \Rightarrow \quad \ln\frac{\alpha\mu_0^2}{2\mu^2} &= -\frac{11}{3}. \end{aligned} \tag{4.92}$$

Using this to eliminate the explicit $\ln(\alpha/2\mu^2)$ dependence,

$$\begin{aligned} \alpha(\rho) &= \alpha + k\alpha^2 \left[3\cdot\frac{11}{3} + 3\ln\frac{\alpha\rho^2}{2\mu^2} \right] \\ &= \alpha + k\alpha^2 \left[-3\ln\frac{\alpha\mu_0^2}{2\mu^2} + 3\ln\frac{\alpha\rho^2}{2\mu^2} \right] \\ &= \alpha + 3k\alpha^2 \ln\left(\frac{\frac{\alpha\rho^2}{2\mu^2}}{\frac{\alpha\mu_0^2}{2\mu^2}} \right), \end{aligned} \tag{4.93}$$

$$\Rightarrow \quad \boxed{\alpha(\rho) = \alpha + \frac{3k\alpha^2}{32\pi^2}\ln\frac{\rho^2}{\mu_0^2},} \tag{4.94}$$

where $\alpha = \alpha(\mu_0)$ is the value of the running coupling at $\rho = \mu_0$. Inverting,

$$\frac{1}{\alpha(\rho)} = \frac{1}{\alpha(\mu_0) + \frac{3k\alpha^2(\mu_0)}{32\pi^2}\ln\frac{\rho^2}{\mu_0^2}} = \frac{1}{\alpha(\mu_0)} \left[1 + \frac{3k\alpha(\mu_0)}{32\pi^2}\ln\frac{\rho^2}{\mu_0^2} \right]^{-1}, \tag{4.95}$$

$$\Rightarrow \quad \frac{1}{\alpha(\rho)} = \frac{1}{\alpha(\mu_0)} \left(1 - \frac{3k\alpha(\mu_0)}{32\pi^2}\ln\frac{\rho^2}{\mu_0^2} \right) + \text{higher order}. \tag{4.96}$$

4.9 Spontaneous Symmetry Breaking by Radiative Corrections?

A natural question arises: can SSB occur in the massless theory purely due to radiative corrections, generating a nonzero $\langle\phi\rangle$ dynamically? From (4.67) with $m_1 = 0$,

$$\begin{aligned} U'(\rho) &= V' + k\,V''V'''\ln\frac{V''}{\mu^2} \\ &= \frac{\alpha}{6}\rho^3 + k\,\frac{\alpha\rho^2}{2}\cdot\alpha\rho\cdot\ln\frac{\alpha\rho^2}{2\mu^2} \\ &= \frac{\rho^3}{6}\left[\alpha + 3k\alpha^2\ln\frac{\alpha\rho^2}{2\mu^2} \right]. \end{aligned} \tag{4.97}$$

Writing $\ln \frac{\alpha\rho^2}{2\mu^2} = \ln \frac{\rho^2}{\mu^2} + \ln \frac{\alpha}{2}$, where the last term is a field-independent constant,

$$U'(\rho) = \frac{\rho^3}{6}\left[\alpha + \frac{3\hbar\alpha^2}{32\pi^2}\left(\ln \frac{\rho^2}{\mu^2} + \text{const}\right)\right]. \tag{4.98}$$

Setting $U'(\langle\phi\rangle) = 0$ for a nontrivial root gives

$$\begin{aligned} & \alpha + \frac{3k\alpha^2}{32\pi^2} \ln \frac{\langle\phi\rangle^2}{\mu^2} = 0 \\ \Rightarrow \quad & \alpha\left(1 + \frac{3k\alpha}{32\pi^2} \ln \frac{\langle\phi\rangle^2}{\mu^2}\right) = 0 \\ \Rightarrow \quad & \hbar\alpha \ln \frac{\langle\phi\rangle^2}{\mu^2} = -\frac{32\pi^2}{3} + \mathcal{O}(\hbar). \end{aligned} \tag{4.99}$$

This root must be rejected. The loop expansion is valid only when $\hbar\alpha$ is small, but (4.99) requires

$$\hbar\alpha \ln \frac{\langle\phi\rangle^2}{\mu^2} \sim \mathcal{O}(1), \tag{4.100}$$

meaning higher-order terms are not small and cannot be neglected. The minimum lies outside the domain of validity of the one-loop approximation. Let us explain this in slightly more details. The loop expansion is an expansion in powers of $\hbar$, with the L-loop contribution carrying a factor of $\hbar^L$. The one-loop approximation is therefore valid when the one-loop correction is small compared to the tree-level term,

$$\frac{\hbar V_1(\rho)}{V(\rho)} \ll 1, \tag{4.101}$$

or more precisely in this case, when

$$\hbar\alpha \ln \frac{\langle\phi\rangle^2}{\mu^2} \ll 1. \tag{4.102}$$

The spurious root requires precisely the opposite, $\hbar\alpha \ln(\langle\phi\rangle^2/\mu^2) \sim \mathcal{O}(1)$, so that the logarithm is of order $1/(\hbar\alpha)$ and $\langle\phi\rangle$ sits exponentially far from μ. At such field values the two-loop contribution goes as

$$(\hbar\alpha)^2 \cdot \left(\ln \frac{\langle\phi\rangle^2}{\mu^2}\right)^2 \sim \mathcal{O}(1), \tag{4.103}$$

and similarly for all higher loops, so every order contributes equally and truncating at one loop is unjustified. The root is therefore rejected, not

of validity of the approximation.

The technical reason for this failure is that the two terms in (4.98) are of order α and α^2 respectively, and cannot cancel each other within the range of validity of perturbation theory: an $\mathcal{O}(\alpha)$ quantity cannot cancel an $\mathcal{O}(\alpha^2)$ quantity. However, if the model were modified so that α and α^2 were replaced by two independent coupling constants, the cancellation could occur and one would obtain a genuine theory of SSB by radiative corrections. The simplest such model is massless scalar electrodynamics, to which we turn in the next section.

Chapter 5

Massless Scalar Electrodynamics and Dimensional Transmutation

In renormalized massless scalar electrodynamics, the scalar field acquires an arbitrary vacuum expectation value $\langle\phi\rangle$ even though classically it vanishes. This means $\langle\phi\rangle$ emerges as a spontaneously generated physical mass scale, with both scalar and vector particles developing dynamical masses proportional to it. This phenomenon is called *dimensional transmutation*: a theory with no classical mass scale develops a preferred mass scale through radiative corrections.

5.1 The Lagrangian and Unitary Gauge

The Lagrangian of massless scalar electrodynamics is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + [(\partial_\mu - ieA_\mu)\phi^*]\,[(\partial^\mu + ieA^\mu)\phi] - \frac{\lambda_0}{6}(\phi^*\phi)^2, \tag{5.1}$$

where e is the renormalized coupling constant and λ_0 is the bare scalar coupling, necessary to renormalize the scalar-scalar scattering amplitude. Transforming away the phase of ϕ by going to unitary gauge and rescaling $\phi \to \phi/\sqrt{2}$, the Lagrangian becomes

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}e^2 A_\mu A^\mu \phi^2 + \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{\lambda_0}{4!}\phi^4, \tag{5.2}$$

subject to the Lorenz condition $\partial_\mu A^\mu = 0$. The corresponding action is

$$S[A,\phi] = \int d^4x \left[\frac{1}{2}A^\mu(\partial^2 + e^2\phi^2)A_\mu - \frac{1}{2}\phi\,\partial^2\phi - \frac{\lambda_0}{4!}\phi^4\right]. \tag{5.3}$$

5.2 Integrating Out the Vector Field

To obtain an effective action for ϕ alone, we integrate out the vector field,

$$\exp iS[\phi] = \mathcal{N}\int(\mathcal{D}A)\,\delta[\partial_\mu A^\mu]\exp iS[A,\phi], \tag{5.4}$$

where $\delta[\partial_\mu A^\mu]$ enforces the Lorenz constraint. Each transverse component of A_μ contributes a Gaussian determinant,

$$\begin{aligned}\left[\det(\partial^2 + e^2\phi^2)\right]^{-\frac{1}{2}} &= \exp\left[-\frac{1}{2}\ln\det(\partial^2 + e^2\phi^2)\right] \\ &= \exp\left[-\frac{1}{2}\mathrm{Tr}\ln(\partial^2 + e^2\phi^2)\right]. \end{aligned} \tag{5.5}$$

There are three such transverse components, so the effective action becomes

$$S[\phi] = -\int d^4x \left(\frac{1}{2}\phi\,\partial^2\phi + \frac{\lambda_0}{4!}\phi^4\right) - \frac{3}{2}\mathrm{Tr}\ln(\partial^2 + e^2\phi^2). \tag{5.6}$$

The one-loop effective potential for ϕ, to lowest order in λ_0, is then

$$U(\rho) = \frac{\lambda_0}{4!}\rho^4 + \frac{3}{2}\int \frac{d^4k_E}{(2\pi)^4}\ln(k_E^2 + e^2\rho^2), \tag{5.7}$$

where the second term sums all one-loop graphs with external ϕ lines and vector boson running in the loop, since we integrated out A_μ. Scalar loops contribute only at higher order.

5.3 The Renormalized Effective Potential

Using the machinery developed in the previous chapter, specifically Eq.(4.65), the renormalized effective potential in this case takes the form

$$U(\rho) = \rho^4\left[\frac{\lambda}{4!} + \frac{3e^4}{64\pi^2}\left(-\frac{1}{2} + \ln\frac{e^2\rho^2}{\mu^2}\right)\right], \tag{5.8}$$

where λ is the running coupling constant at the renormalization point μ. Note that e^4 is a one-loop result and λ is a tree-level result, but they can be comparable since they are independent coupling constants. This is what allows the cancellation that was impossible in the pure scalar case.

5.4 Spontaneous Symmetry Breaking and Particle Masses

Differentiating (5.8),

$$\begin{aligned} U'(\rho) &= 4\rho^3\left[\cdots\right] + \rho^4\cdot\frac{3e^4}{64\pi^2}\cdot\frac{1}{\frac{e^2\rho^2}{\mu^2}}\cdot\frac{2\rho e^2}{\mu^2} \\ &= 4\rho^3\left[\cdots\right] + \frac{3e^4}{64\pi^2}\rho^3\cdot 2 \\ &= \rho^3\left[4\cdot\frac{\lambda}{4!} + 4\cdot\frac{3e^4}{64\pi^2}\left(-\frac{1}{2}\right) + 4\cdot\frac{3e^4}{64\pi^2}\ln\frac{e^2\rho^2}{\mu^2} + \frac{3e^4}{64\pi^2}\cdot 2\right], \end{aligned} \tag{5.9}$$

$$\Rightarrow U'(\rho) = 4\rho^3\left[\frac{\lambda}{4!} + \frac{3e^4}{64\pi^2}\ln\frac{e^2\rho^2}{\mu^2}\right]. \tag{5.10}$$

Setting $U'(\langle\phi\rangle) = 0$ for a nontrivial root,

$$\begin{aligned} \frac{\lambda}{4!} + \frac{3e^4}{64\pi^2} \ln \frac{e^2\langle\phi\rangle^2}{\mu^2} &= 0 \\ \Rightarrow \quad \frac{3e^4}{64\pi^2} \ln \frac{e^2\langle\phi\rangle^2}{\mu^2} &= -\frac{\lambda}{4!}. \end{aligned} \tag{5.11}$$

Unlike the pure scalar case, this root is within the domain of validity of the approximation since λ and e^4 are independent and can be chosen to make the two terms comparable without requiring the logarithm to be large. Using this condition to eliminate μ from (5.8),

$$\begin{aligned} U(\rho) &= \rho^4 \left[-\frac{3e^4}{64\pi^2} \ln \frac{e^2\langle\phi\rangle^2}{\mu^2} + \frac{3e^4}{64\pi^2}\left(-\frac{1}{2}\right) + \frac{3e^4}{64\pi^2} \ln \frac{e^2\rho^2}{\mu^2} \right] \\ &= \frac{3e^4\rho^4}{64\pi^2}\left[-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2}\right]. \end{aligned} \tag{5.12}$$

The scalar particle mass follows from $m_S^2 \equiv U''(\langle\phi\rangle)$. Setting $a \equiv 3e^4/(64\pi^2)$ for brevity, we differentiate $U(\rho) = a\rho^4\left[-\frac{1}{2} + \ln\frac{\rho^2}{\langle\phi\rangle^2}\right]$,

$$\begin{aligned} U'(\rho) &= a \cdot 4\rho^3 \left[-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2}\right] + a\rho^4 \cdot \frac{\langle\phi\rangle^2}{\rho^2} \cdot \frac{2\rho}{\langle\phi\rangle^2} \\ &= 2a\rho^3 \left[2\left(-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2}\right) + 1\right] \\ &= 4a\rho^3 \ln \frac{\rho^2}{\langle\phi\rangle^2}. \end{aligned} \tag{5.13}$$

Differentiating once more,

$$\begin{aligned} U''(\rho) &= 12a\rho^2 \ln \frac{\rho^2}{\langle\phi\rangle^2} + 4a\rho^3 \cdot \frac{\langle\phi\rangle^2}{\rho^2} \cdot \frac{2\rho}{\langle\phi\rangle^2} \\ &= 12a\rho^2 \ln \frac{\rho^2}{\langle\phi\rangle^2} + 8a\rho^2. \end{aligned} \tag{5.14}$$

Evaluating at $\rho = \langle\phi\rangle$,

$$\begin{aligned} U''(\langle\phi\rangle) &= 12a\langle\phi\rangle^2 \ln \frac{\langle\phi\rangle^2}{\langle\phi\rangle^2} + 8a\langle\phi\rangle^2 \\ &= 8 \cdot \frac{3e^4}{64\pi^2}\langle\phi\rangle^2, \end{aligned} \tag{5.15}$$

$$\Rightarrow \quad \boxed{m_S^2 = \frac{3e^4}{8\pi^2}\langle\phi\rangle^2.} \tag{5.16}$$

The vector field acquires mass through the Higgs mechanism,

$$m_V^2 = e^2\langle\phi\rangle^2. \tag{5.17}$$

Both masses are proportional to $\langle\phi\rangle$, confirming that the dynamically generated VEV sets the mass scale of the theory.

5.5 Dimensional Transmutation

To renormalize the theory we introduce a momentum cutoff, which represents a hidden scale. After renormalization this hidden scale is replaced by an arbitrary finite renormalization point μ. In a classically massless theory one might expect all choices of μ to be equivalent, but what we have found is that there is a preferred renormalization point: the one at which the particles develop mass dynamically. Dimensional transmutation refers precisely to this emergence of a preferred scale in a theory with no intrinsic mass, whereby the infinite cutoff momentum becomes transmuted into a finite physical mass. This mechanism is widely believed to operate in QCD with massless quarks, where no intrinsic mass scale exists in the Lagrangian yet hadrons acquire mass dynamically.

Chapter 6

The Glashow-Salam-Weinberg Model

The Glashow-Salam-Weinberg (GSW) model is the quantum field theory of electroweak interactions, unifying the electromagnetic and weak forces into a single gauge theory with gauge group $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. It is one of the cornerstones of the Standard Model and provides the first physical example of spontaneous symmetry breaking in a gauge theory, the Higgs mechanism, which we studied abstractly in the previous chapter. Here we apply the effective potential formalism to this model, computing the field-dependent gauge boson masses and the resulting one-loop correction to the effective potential.

6.1 The Electroweak Lagrangian

The gauge fields of the theory are the $\mathrm{SU}(2)_L$ triplet W^i_μ ($i = 1, 2, 3$) and the $\mathrm{U}(1)_Y$ singlet V_μ, with corresponding field strengths $W^i_{\mu\nu}$ and $V_{\mu\nu}$. The matter content includes left-handed fermion doublets and right-handed singlets, but for the purposes of computing the effective potential it is the scalar sector that is relevant. The Higgs field φ is a complex doublet under $\mathrm{SU}(2)_L$ with hypercharge $Y = 1$. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4} W^i_{\mu\nu} W^{i\,\mu\nu} - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + (D_\mu \varphi)^\dagger (D^\mu \varphi) - V(\varphi), \tag{6.1}$$

where the covariant derivative is

$$D_\mu = \partial_\mu - ig\frac{\sigma^i}{2} W^i_\mu - ig'\frac{Y}{2} V_\mu, \tag{6.2}$$

with g and g' the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ coupling constants respectively, and σ^i the Pauli matrices. The classical potential $V(\varphi) = -\mu^2 \varphi^\dagger \varphi + \lambda(\varphi^\dagger \varphi)^2$ has a minimum at a nonzero field value when $\mu^2 > 0$, triggering spontaneous symmetry breaking.

6.2 Spontaneous Symmetry Breaking and Mass Generation

The Higgs field acquires a vacuum expectation value

$$\langle \varphi \rangle = \begin{pmatrix} 0 \\ a \end{pmatrix}, \tag{6.3}$$

which breaks $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ down to the electromagnetic $\mathrm{U}(1)_{\mathrm{em}}$. The gauge boson masses are read off from the kinetic term evaluated at the VEV. Since the VEV is constant, the ∂_μ term vanishes and

$$D_\mu \langle\varphi\rangle = \left(-ig\frac{\sigma^i}{2}W_\mu^i - ig'\frac{Y}{2}V_\mu\right)\begin{pmatrix}0\\a\end{pmatrix}. \tag{6.4}$$

Expanding the SU(2) part using

$$\sigma^i W_\mu^i = \begin{pmatrix} W_\mu^3 & W_\mu^1 - iW_\mu^2 \\ W_\mu^1 + iW_\mu^2 & -W_\mu^3 \end{pmatrix}, \tag{6.5}$$

acting on the VEV gives

$$\frac{\sigma^i}{2}W_\mu^i\begin{pmatrix}0\\a\end{pmatrix} = \frac{a}{2}\begin{pmatrix}W_\mu^1 - iW_\mu^2\\ -W_\mu^3\end{pmatrix}. \tag{6.6}$$

For the U(1) part, with hypercharge $Y = 1$,

$$\frac{Y}{2}V_\mu\begin{pmatrix}0\\a\end{pmatrix} = \frac{a}{2}\begin{pmatrix}0\\V_\mu\end{pmatrix}. \tag{6.7}$$

Combining,

$$D_\mu\langle\varphi\rangle = -ig\cdot\frac{a}{2}\begin{pmatrix}W_\mu^1 - iW_\mu^2\\ -W_\mu^3\end{pmatrix} - ig'\cdot\frac{a}{2}\begin{pmatrix}0\\V_\mu\end{pmatrix} = \frac{ia}{2}\begin{pmatrix}gW_\mu^1 - igW_\mu^2\\ -gW_\mu^3 + g'V_\mu\end{pmatrix}. \tag{6.8}$$

Taking the conjugate,

$$\overline{D_\mu\langle\phi\rangle} = -\frac{ia}{2}\left(gW_\mu^1 + igW_\mu^2,\ -gW_\mu^3 + g'V_\mu\right), \tag{6.9}$$

so that

$$\overline{D_\mu\langle\phi\rangle}\, D^\mu\langle\phi\rangle = \frac{a^2}{4}\left(gW_\mu^1 + igW_\mu^2,\ -gW_\mu^3 + g'V_\mu\right)\begin{pmatrix}gW^{1\mu} - igW^{2\mu}\\ -gW^{3\mu} + g'V^\mu\end{pmatrix}. \tag{6.10}$$

Expanding the matrix product,

$$= \frac{a^2}{4}\left[g^2(W_\mu^{1\,2} + W_\mu^{2\,2}) + g^2W_\mu^{3\,2} + g'^2V_\mu^2 - 2gg'W_\mu^3V_\mu\right]. \tag{6.11}$$

The fields W_μ^3 and V_μ mix, and to identify the physical mass eigenstates we introduce the Weinberg rotation,

$$A_\mu = \sin\theta_W\, W_\mu^3 + \cos\theta_W\, V_\mu, \tag{6.12}$$
$$Z_\mu = \cos\theta_W\, W_\mu^3 - \sin\theta_W\, V_\mu, \tag{6.13}$$

where $\sin\theta_W = g'/\sqrt{g^2+g'^2}$ and $\cos\theta_W = g/\sqrt{g^2+g'^2}$. Inverting,

$$W_\mu^3 = \sin\theta_W\, A_\mu + \cos\theta_W\, Z_\mu, \tag{6.14}$$
$$V_\mu = \cos\theta_W\, A_\mu - \sin\theta_W\, Z_\mu. \tag{6.15}$$

Substituting into each term,

$$\begin{aligned} g^2 W_\mu^{3\,2} &= g^2 \sin^2\theta_W A_\mu^2 + g^2\cos^2\theta_W Z_\mu^2 + 2g^2 \sin\theta_W\cos\theta_W A_\mu Z_\mu, \\ g'^2 V_\mu^2 &= g'^2\cos^2\theta_W A_\mu^2 + g'^2 \sin^2\theta_W Z_\mu^2 - 2g'^2 \sin\theta_W\cos\theta_W A_\mu Z_\mu, \\ -2gg' W_\mu^3 V_\mu &= -2gg'(\sin\theta_W A_\mu + \cos\theta_W Z_\mu)(\cos\theta_W A_\mu - \sin\theta_W Z_\mu) \\ = -2gg'\sin\theta_W\cos\theta_W A_\mu^2 &+ 2gg'\sin^2\theta_W A_\mu Z_\mu - 2gg'\cos^2\theta_W Z_\mu A^\mu + 2gg'\cos\theta_W\sin\theta_W Z_\mu^2. \end{aligned} \tag{6.16}$$

Collecting by field bilinear,

$$\begin{aligned} g^2 W_\mu^{3\,2} + g'^2 V_\mu^2 - 2gg' W_\mu^3 V^\mu &= A_\mu^2 \left(g\sin\theta_W - g'\cos\theta_W\right)^2 \\ &+ Z_\mu^2 \left(g\cos\theta_W + g'\sin\theta_W\right)^2 \\ &+ 2\sin\theta_W\cos\theta_W A_\mu Z^\mu \left(g^2 - g'^2 + gg'\left(\frac{\sin\theta_W}{\cos\theta_W} - \frac{\cos\theta_W}{\sin\theta_W}\right)\right). \end{aligned} \tag{6.17}$$

The coefficient of A_μ^2 vanishes identically,

$$g\sin\theta_W - g'\cos\theta_W = \frac{gg'}{\sqrt{g^2+g'^2}} - \frac{g'g}{\sqrt{g^2+g'^2}} = 0, \tag{6.18}$$

and the $A_\mu Z^\mu$ cross term also vanishes. The coefficient of Z_μ^2 evaluates to

$$\left(g\cos\theta_W + g'\sin\theta_W\right)^2 = \left(\frac{g^2}{\sqrt{g^2+g'^2}} + \frac{g'^2}{\sqrt{g^2+g'^2}}\right)^2 = g^2 + g'^2. \tag{6.19}$$

The photon therefore remains massless, as required by the unbroken $\mathrm{U}(1)_{\mathrm{em}}$, and the kinetic term becomes

$$\boxed{\overline{D_\mu\langle\varphi\rangle}\, D^\mu\langle\varphi\rangle = \frac{a^2}{4}\left[g^2\left(W_\mu^{1\,2} + W_\mu^{2\,2}\right) + \left(g^2+g'^2\right) Z_\mu^2\right].} \tag{6.20}$$

Reading off the field-dependent masses,

$$m^2(W_{1,2}) = \frac{1}{4} g^2 a^2, \tag{6.21}$$

$$m^2(Z) = \frac{1}{4}(g^2+g'^2)a^2, \tag{6.22}$$

while the photon remains massless. These are the standard results of the Higgs mechanism in the electroweak theory.

6.3 One-Loop Gauge Field Contribution to the Effective Potential

With the field-dependent gauge boson masses identified, we can compute their contribution to the one-loop effective potential. By the same logic as in the scalar case, each massive gauge boson with field-dependent mass matrix M^2 contributes a Coleman-Weinberg type term at one loop. The gauge field contribution is

$$V_g^{(1)}(\bar{\varphi}) = \frac{3}{64\pi^2}\,\mathrm{Tr}\big(M^4 \ln M^2\big)\Big|_{\rho=\bar{\varphi}}, \qquad M^2_{\alpha\beta} = \lambda_\alpha\lambda_\beta\,\mathrm{Tr}\big[(t_\alpha\varphi)^\dagger\tau_\beta\varphi\big]\,, \tag{6.23}$$

where the factor of 3 accounts for the three physical polarizations of each massive gauge boson. Substituting the masses of the two W bosons and the Z boson,

$$\begin{aligned} V_g^{(1)}(\bar{\varphi}) &= \frac{3}{64\pi^2}\left(\frac{1}{16}g^4 + \frac{1}{16}g^4 + \frac{1}{16}(g^2+g'^2)^2\right)\bar{\varphi}^4 \ln \bar{\varphi}^2 \\ &= \frac{3}{64\pi^2}\cdot\frac{1}{16}\left[2g^4 + (g^2+g'^2)^2\right]\bar{\varphi}^4 \ln\bar{\varphi}^2, \end{aligned} \tag{6.24}$$

$$\boxed{\Rightarrow \quad V_g^{(1)}(\bar{\varphi}) = \frac{3}{1024\pi^2}\left[2g^4 + (g^2+g'^2)^2\right]\bar{\varphi}^4 \ln\bar{\varphi}^2.} \tag{6.25}$$

The coefficient is determined entirely by the gauge couplings g and g', and the structure $\bar{\varphi}^4 \ln\bar{\varphi}^2$ is the characteristic signature of the Coleman-Weinberg mechanism in a gauge theory. In the next chapter we extend this analysis to GUT models, where the larger gauge group and richer symmetry breaking pattern lead to qualitatively new features.

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Appendix A

Proof that $Z(J) = e^{iW(J)}$

The connection between $Z(J)$ and $W(J)$ follows from the combinatorics of Wick diagrams. The S-matrix is

$$S = T \exp\left[-i \int_{-\infty}^{\infty} dt\, H_I(t)\right], \tag{A.1}$$

where H_I is the interaction picture Hamiltonian, which here includes a source term. Expanding S in a Taylor series and applying Wick's theorem gives a sum over Wick diagrams,

$$S = \sum \mathcal{W}, \tag{A.2}$$

where each $\mathcal{W}$ is a Wick graph. For example, for an interaction of the form $H_I(x) = \lambda\bar{\psi}(x)\psi(x)\varphi(x)$, one of the terms at order λ^2 looks like

Unlike Feynman diagrams, Wick diagrams are labelled. Taking the vacuum expectation value of S selects only diagrams with no uncontracted external legs,

$$Z(J) = \langle 0|S|0\rangle = \sum \mathcal{W}_u, \tag{A.3}$$

where $\mathcal{W}_u$ denotes vacuum graphs. A vacuum graph is defined as a graph with no external legs, only internal lines forming closed loops,

Any collection of vacuum graphs combined into a single disconnected diagram is still a vacuum graph, since no external legs appear anywhere. Similarly, $iW(J) = \sum \mathcal{W}_{cu}$, where $\mathcal{W}_{cu}$ are the connected vacuum graphs. It then follows from the theorem proved in Appendix B that

$$Z(J) = \sum \mathcal{W}_u = \exp \sum \mathcal{W}_{cu} = e^{iW(J)}, \tag{A.4}$$

which completes the proof. $\square$

Appendix B

The Sum of All Wick Diagrams is the Exponential of the Connected Ones

We prove that

$$\sum \text{ all Wick diagrams} = \exp\Big(\sum \text{ connected Wick diagrams}\Big) . \tag{B.1}$$

Consider the case where H_I contains a single interaction term. Two diagrams belong to the same pattern P if they differ only by a permutation of vertices. Define $S(P)$ as the number of permutations that leave a diagram of pattern P unchanged, and $n(P)$ as the number of vertices. Then the S-matrix is

$$S = \sum_P \frac{:\mathcal{O}(P):}{S(P)}, \tag{1}$$

where $:\mathcal{O}(P):$ is the normal-ordered operator represented by the Wick diagram. Now consider a pattern consisting of n_i copies of the connected pattern P_i, with $i = 1, 2, \ldots$ an ordering of all connected patterns. Then

$$\mathcal{O}(P) = \prod_{i=1}^{\infty} \mathcal{O}(P_i)^{n_i}, \tag{2}$$

$$S(P) = \prod_{i=1}^{\infty} S(P_i)^{n_i}. \tag{3}$$

Substituting into (1), the additional $n_i!$ in the denominator accounts for the fact that permuting graphs of a given connected pattern does not yield a new Wick graph,

$$S = \sum_{\substack{n_i=0 \\ i=0}}^{\infty} \frac{:\prod_{i=1}^{\infty} \mathcal{O}(P_i)^{n_i}:}{\prod_{i=1}^{\infty} \left[S(P_i)^{n_i}\, n_i!\right]}. \tag{4}$$

Factoring the sum over each n_i independently,

$$\begin{aligned}
S &= \prod_{i=1}^{\infty} \sum_{n_i=0}^{\infty} \frac{1}{n_i!} \left[\frac{:\mathcal{O}(P_i):}{S(P_i)} \right]^{n_i} \\
&=: \prod_{i=1}^{\infty} \exp\left(\frac{\mathcal{O}(P_i)}{S(P_i)} \right) : \\
&=: \exp\left(\sum_{i=1}^{\infty} \frac{\mathcal{O}(P_i)}{S(P_i)} \right) : \\
&=: \exp \sum \mathcal{W}_c :,
\end{aligned} \tag{5}$$

where $\mathcal{W}_c$ denotes connected Wick graphs and we used (2) and (3) in the last step. This establishes that

$$\sum \text{all Wick diagrams} = e^{\sum \text{ connected Wick diagrams}}. \tag{6}$$

$\square$

The proof of $Z(J) = e^{iW(J)}$ in Appendix A then follows immediately by taking the vacuum expectation value of (6).

Appendix C

Equivalence of the Loop Expansion and the $\hbar$ Expansion

We show that an L-loop graph carries a factor of $\hbar^{L-1}$, so that the loop expansion and the $\hbar$ expansion are equivalent. Starting from

$$Z(J) = N \int [\mathcal{D}\varphi] \exp\left[\frac{i}{\hbar} S(\varphi, J)\right], \tag{C.1}$$

the vertex factors are the coefficients of the interaction terms in $\hbar^{-1}S(\varphi, J)$, so each vertex contributes a factor of $\hbar^{-1}$. The propagators are the inverses of the quadratic operators in the exponent, so each internal line contributes a factor of $\hbar$. A graph with V vertices and I internal lines therefore carries an overall factor of

$$\hbar^{I-V}. \tag{C.2}$$

For a connected graph, each vertex fixes one internal momentum except for one overall energy-momentum conservation constraint, so the number of independent loop momenta is

$$L = I - V + 1. \tag{C.3}$$

Hence the $\hbar$-dependence of a connected graph is

$$\hbar^{I-V} = \hbar^{L-1}, \tag{C.4}$$

which shows that the loop expansion is precisely the expansion in powers of $\hbar$. $\square$

Appendix D

Derivation of $\delta W(J)/\delta J = \varphi$

This appendix supplies the step used in the proof of Theorem 3, following Jackiw and Kerman (1979). The two equivalent expressions for $W(J)$ are

$$e^{\frac{i}{\hbar}W(J)} = \langle -, t | +, t \rangle, \tag{D.1}$$

$$e^{\frac{i}{\hbar}W(J)} = \langle 0 | \, T \exp\left(\frac{i}{\hbar} \int (dx) \, J(x) \, \Phi(x) \right) | 0 \rangle. \tag{D.2}$$

The states $|\pm, t\rangle$ solve the time-dependent Schrödinger equation

$$i\hbar \, \partial_t |\pm, t\rangle = H |\pm, t\rangle - \int d\mathbf{x} \, J(t, \mathbf{x}) \, \Phi(\mathbf{x}) |\pm, t\rangle, \tag{D.3}$$

with boundary conditions $\lim_{t \to \mp\infty} |\pm, t\rangle = |0\rangle$, where $|0\rangle$ is the ground state of H with $H|0\rangle = 0$. The states $|\psi_\pm, t\rangle$ are related to $|\pm, t\rangle$ by

$$|+, t\rangle = \exp \frac{i}{\hbar} \left(\int_{-\infty}^{t} dt' \, w(t') \right) |\psi_+, t\rangle, \tag{D.4}$$

$$|-, t\rangle = \exp \frac{i}{\hbar} \left(- \int_{t}^{\infty} dt' \, w^*(t') \right) |\psi_-, t\rangle, \tag{D.5}$$

and satisfy $\left(i\hbar \, \partial_t - H + \int J\Phi \, dx \right) |\pm, t\rangle = 0$ with the same boundary conditions. Their overlap gives

$$\langle -, t | +, t \rangle = e^{\frac{i}{\hbar}W(J)}. \tag{D.6}$$

Expanding this explicitly,

$$\begin{aligned}
&\langle \psi_-, t | \left[\exp \frac{i}{\hbar} \left(- \int_{t}^{\infty} dt' \, w^*(t') \right) \right]^* \exp \frac{i}{\hbar} \left(\int_{-\infty}^{t} w(t') \, dt' \right) |\psi_+, t\rangle \\
&= \langle \psi_-, t | \exp \frac{i}{\hbar} \left(\int_{t}^{\infty} dt' \, w(t') + \int_{-\infty}^{t} dt' \, w(t') \right) |\psi_+, t\rangle \\
&= \langle \psi_-, t | \exp \frac{i}{\hbar} \int_{-\infty}^{\infty} dt' \, w(t') \, |\psi_+, t\rangle \\
&= \exp \frac{i}{\hbar} \int_{-\infty}^{\infty} dt' \, w(t') \, \underbrace{\langle \psi_-, t | \psi_+, t \rangle}_{1} \\
&= \exp \frac{i}{\hbar} \left(\int_{-\infty}^{\infty} dt' \, w(t') \right),
\end{aligned} \tag{D.7}$$

so that $W = \int_{-\infty}^{\infty} dt\, w(t)$. Using the eigenvalue equation for $|\psi_+, t\rangle$,

$$\int_{-\infty}^{\infty} dt\, \langle \psi_-, t| \left(i\hbar\, \partial_t - H + \int d\mathbf{x}\, J\Phi \right) |\psi_+, t\rangle = \int_{-\infty}^{\infty} dt\, w(t)\, \underbrace{\langle \psi_-, t|\psi_+, t\rangle}_{1}, \tag{D.8}$$

which gives

$$W(J) = \int_{-\infty}^{\infty} dt\, \langle \psi_-, t|\, i\hbar\, \partial_t - H + \int d\mathbf{x}\, J\Phi\, |\psi_+, t\rangle. \tag{D.9}$$

Taking the functional derivative,

$$\begin{aligned} \frac{\delta W(J)}{\delta J} &= \int_{-\infty}^{\infty} dt\, \frac{\delta \langle \psi_-, t|}{\delta J} \underbrace{\left(i\hbar\, \partial_t - H + \int J\Phi \right) |\psi_+, t\rangle}_{w(t)|\psi_+,t\rangle} \\ &+ \int_{-\infty}^{\infty} dt\, \langle \psi_-, t| \frac{\delta}{\delta J} \left(i\hbar\, \partial_t - H + \int d\mathbf{x}\, J\Phi \right) |\psi_+, t\rangle \\ &+ \int_{-\infty}^{\infty} dt\, \langle \psi_-, t| \left(i\hbar\, \partial_t - H + \int d\mathbf{x}\, J\Phi \right) \frac{\delta |\psi_+, t\rangle}{\delta J}, \end{aligned} \tag{D.10}$$

where the constraint $\langle \psi_-, t|\Phi(\mathbf{x})|\psi_+, t\rangle = \phi(t, \mathbf{x})$ was used for the second term. Integrating the third term by parts,

$$i\hbar \left\langle \psi_-, t \middle| \frac{\delta |\psi_+, t\rangle}{\delta J} \right\rangle \Bigg|_{-\infty}^{\infty} + \left\langle \psi_-, t \middle| \underbrace{\left(-i\hbar\, \partial_t - H + \int d\mathbf{x}\, J\Phi \right)}_{\langle \psi_-, t| w(t)} \frac{\delta |\psi_+, t\rangle}{\delta J} \right\rangle, \tag{D.11}$$

so that

$$\begin{aligned} \frac{\delta W(J)}{\delta J} &= \phi + i\hbar \left\langle \psi_-, t \middle| \frac{\delta}{\delta J} |\psi_+, t\rangle \right\rangle \Bigg|_{-\infty}^{\infty} \\ &+ \int_{-\infty}^{\infty} dt\, w(t) \left[\frac{\delta \langle \psi_-, t|}{\delta J} |\psi_+, t\rangle + \langle \psi_-, t| \frac{\delta}{\delta J} |\psi_+, t\rangle \right], \end{aligned} \tag{D.12}$$

where the expression inside the brackets vanishes upon differentiating the normalization constraint $\langle \psi_-, t|\psi_+, t\rangle = 1$. It remains to show that the boundary term also vanishes. At $t \to -\infty$, $|\psi_+, t\rangle \to |0\rangle$ independently of J, so $\delta|\psi_+, t\rangle/\delta J = 0$ at the lower limit. For the upper limit, rewrite it as

$$i\hbar \frac{\delta}{\delta J} \underbrace{\langle \psi_-, t|\psi_+, t\rangle}_{1} - i\hbar \left(\frac{\delta \langle \psi_-, t|}{\delta J} \right) |\psi_+, t\rangle, \tag{D.13}$$

where the second term vanishes because at $t \to \infty$, $\langle \psi_-, t| \to \langle 0|$, also independently of J. The boundary term therefore vanishes at both limits, and

$$\therefore \quad \boxed{\frac{\delta W(J)}{\delta J} = \phi.} \tag{D.14}$$

Appendix E

Matrix Element of the Inverse Propagator

We evaluate $\langle k'|\, i\Delta^{-1}[\rho]\,|k\rangle$ using $\langle x|k\rangle = e^{ikx}$,

$$\begin{aligned}
\langle k'|\, i\Delta^{-1}[\rho]\,|k\rangle &= \int d^4x\, d^4y\ \langle k'|x\rangle\langle x|\, i\Delta^{-1}[\rho]\,|y\rangle\langle y|k\rangle \\
&= \int d^4x\, d^4y\ e^{ik'x}\left[-\left(\Box + V''(\rho)\right)\right]\delta^4(x-y)\, e^{iky} \\
&= \int d^4x\ e^{-i(k-k')x}\left[-(-k^2 + V''(\rho))\right] \\
&= \left[k^2 - V''(\rho)\right](2\pi)^4\delta^4(k-k'). \qquad \text{(E.1)}
\end{aligned}$$

Setting $k = k'$ and dividing by $\Omega = (2\pi)^4\delta^4(0)$,

$$\langle k|\, i\Delta^{-1}(\rho)\,|k\rangle = \left[k^2 - V''(\rho)\right]\Omega, \qquad \text{(E.2)}$$

$$\Rightarrow \quad \Omega^{-1}\langle k|\, i\Delta^{-1}(\rho)\,|k\rangle = k^2 - V''(\rho). \qquad \text{(E.3)}$$

Appendix F

Evaluation of the One-Loop Momentum Integral

We evaluate

$$I = \int \frac{d^4k_E}{(2\pi)^4} \ln(k_E^2 + V'') = \frac{1}{8\pi^2} \int_0^\Lambda dk_E\, k_E^3 \ln(k_E^2 + V''), \tag{F.1}$$

where the angular integration in four-dimensional Euclidean space gives a factor $2\pi^2$. Substituting $u = k_E^2$, $du = 2k_E\, dk_E$,

$$I = \frac{1}{16\pi^2} \int_0^{\Lambda^2} u\, du \ln(u + V''). \tag{F.2}$$

Integrating by parts with $f = \ln(u + V'')$, $dg = u\, du$,

$$I = \frac{1}{16\pi^2} \left\{ \left[\frac{u^2}{2} \ln(u + V'') \right]_0^{\Lambda^2} - \int_0^{\Lambda^2} \frac{u^2}{2} \cdot \frac{1}{u + V''}\, du \right\}, \tag{F.3}$$

where the boundary term evaluates to $\dfrac{\Lambda^4}{2} \ln(\Lambda^2 + V'')$. Using the partial fraction decomposition $\dfrac{u^2}{u + V''} = u - V'' + \dfrac{(V'')^2}{u + V''}$,

$$\begin{aligned}
\frac{1}{2} \int_0^{\Lambda^2} \frac{u^2}{u + V''}\, du &= \frac{1}{2} \int_0^{\Lambda^2} \left(u - V'' + \frac{(V'')^2}{u + V''} \right) du \\
&= \frac{1}{2} \left[\frac{\Lambda^4}{2} - \Lambda^2 V'' + (V'')^2 \ln(\Lambda^2 + V'') - (V'')^2 \ln V'' \right].
\end{aligned} \tag{F.4}$$

Assembling the pieces,

$$I = \frac{1}{16\pi^2} \left\{ \frac{\Lambda^4}{2} \ln(\Lambda^2 + V'') - \frac{1}{2} \left[\frac{\Lambda^4}{2} - \Lambda^2 V'' + (V'')^2 \ln\left(\frac{\Lambda^2 + V''}{V''} \right) \right] \right\}. \tag{F.5}$$

Expanding $\ln(\Lambda^2 + V'') = \ln \Lambda^2 + \ln(1 + V''/\Lambda^2) \simeq \ln \Lambda^2 + V''/\Lambda^2 - (V'')^2/\Lambda^4 + \cdots$, the leading terms are

$$\frac{\Lambda^4}{2} \ln(\Lambda^2 + V'') \simeq \Lambda^4 \ln \Lambda + \frac{\Lambda^2 V''}{2} - \frac{(V'')^2}{2}, \tag{F.6}$$

$$\frac{(V'')^2}{2} \ln(\Lambda^2 + V'') \simeq \frac{(V'')^2}{2} \ln \Lambda^2 + \mathcal{O}(1/\Lambda^2). \tag{F.7}$$

Collecting all terms,

$$I \quad = \quad \frac{1}{16\pi^2}\left\{\Lambda^4\left(\ln\Lambda - \frac{1}{4}\right) + \Lambda^2 V'' + \frac{(V'')^2}{2}\left(\ln\frac{\Lambda^2}{V''} - \frac{1}{2}\right)\right\} \; + \; \mathcal{O}\left(\frac{1}{\Lambda^2}\right). \quad \text{(F.8)}$$

Dropping the $\Lambda^n \ln\Lambda$ terms, which are field-independent and do not affect the location of the minimum of $U(\rho)$, the one-loop integral contributing to (4.49) is

$$\frac{\hbar}{2}I \quad = \quad \hbar\left[\frac{\Lambda^2}{32\pi^2}V''(\rho) + \frac{[V''(\rho)]^2}{64\pi^2}\left(-\frac{1}{2} + \ln\frac{V''(\rho)}{\Lambda^2}\right) + \mathcal{O}\left(\frac{1}{\Lambda^2}\right)\right]. \quad \text{(F.9)}$$

Bibliography